arXiv:2509.21869v3 [math.CA] 22 Jun 2026

TWO-ENDS FURSTENBERG ESTIMATES IN THE PLANE

HONG WANG AND SHUKUN WU

Abstract. We prove two-ends Furstenberg estimates in the plane for a Katz–Tao (δ, t)-set of lines, for general $t \in [0, 2]$.

1. Introduction

In this paper, we extend the two-ends Furstenberg inequality established in [WW24] to a broader class of lines. We begin with a standard definition.

Definition 1.1 (Shading)**.** *Let L be a family of lines in $\mathbb{R}^2$ and let $\delta \in (0, 1)$. A* **shading** *$Y : L \to B^2(0, 1)$ is an assignment such that $Y(\ell) \subset N_\delta(\ell) \cap B^2(0, 1)$ is a union of δ-balls in $\mathbb{R}^2$ for all $\ell \in L$. We write $(L, Y)_\delta$ to emphasize the dependence on δ. A similar definition applies to a family of δ-tubes $\mathcal{T}$.*

Regarding the distribution of a shading $Y(\ell)$, the next definition gives a quantitative "two-ends" condition, which is a fairly weak non-concentration condition.

Definition 1.2 (Two-ends)**.** *Let $\delta \in (0, 1)$ and let $(L, Y)_\delta$ be a set of lines and shading. Let $0 < \varepsilon_2 < \varepsilon_1 < 1$. We say Y is $(\varepsilon_1, \varepsilon_2, C)$-***two-ends*** if for all $\ell \in L$ and all $\delta \times \delta^{\varepsilon_1}$-tubes $J \subset N_\delta(\ell)$,*

$$|Y(\ell) \cap J| \leqslant C\delta^{\varepsilon_2}|Y(\ell)|. \tag{1.1}$$

*When the constant C is not important in the context, we say Y is $(\varepsilon_1, \varepsilon_2)$-***two-ends***, or simply* **two-ends***. A similar definition applies to a single shading $Y(\ell)$.*

Now let us recall [WW24, Theorem 2.1]:

Theorem 1.3. *Let $\delta \in (0, 1)$. Let $(L, Y)_\delta$ be a set of directional δ-separated lines in $\mathbb{R}^2$ with an $(\varepsilon_1, \varepsilon_2)$-two-ends shading such that $|Y(\ell)| \geqslant \lambda\delta$ for all $\ell \in L$. Then for all $\varepsilon > 0$,*

$$\Big|\bigcup_{\ell\in L} Y(\ell)\Big| \geqslant c_\varepsilon \delta^\varepsilon \delta^{\varepsilon_1/2} \lambda^{1/2} \sum_{\ell\in L} |Y(\ell)|. \tag{1.2}$$

As discussed in [WW24, Section 0.3], (1.2) matches the numerology of the Furstenberg set inequality established in [RW23, Theorem 4.1] (see Theorem 3.6). The assumption that L is a set of directional δ-separated lines can easily be replaced by the superficially stronger assumption that L is a Katz–Tao $(\delta, 1)$-set of lines. The main result of this paper is a generalization of Theorem 1.3 to a set of lines and two-ends shading $(L, Y)_\delta$, where L is a Katz–Tao (δ, t)-set of lines for general $t \in [0, 2]$ (see Definition 2.5 for the definition of a Katz–Tao (δ, t)-set of lines). We begin with a necessary definition, which is essentially the Katz–Tao constant introduced in Definition 2.4.

Definition 1.4. *Let $\delta \in (0,1]$, and let $(L,Y)_\delta$ be a set of lines and shading. Let $t \in [0,1]$. For each $\ell \in L$, define*

$$\gamma_{Y,t}(\ell) := \sup_{r\in[\delta,1],\, x\in\ell} (\delta/r)^t(\delta^{-2}|B(x,r)\cap Y(\ell)|), \tag{1.3}$$

and define $\gamma_{Y,t} := \sup_{\ell\in L}\gamma_{Y,t}(\ell)$.

Theorem 1.5. *Let $(L,Y)_\delta$ be a set of δ-separated lines in $\mathbb{R}^2$ with a $(\varepsilon_1,\varepsilon_2)$-two-ends shading such that $|Y(\ell)| \geqslant \lambda\delta$ for all $\ell \in L$. Let $t \in (0,2)$, and $t^* = \min\{t, 2-t\}$. Suppose the set of lines L is a Katz–Tao (δ,t)-set. Then for all $\varepsilon > 0$,*

$$\Big|\bigcup_{\ell\in L} Y(\ell)\Big| \geqslant c_\varepsilon \delta^\varepsilon \delta^{t\varepsilon_1/2}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t^*}^{-1/2}\sum_{\ell\in L}|Y(\ell)|. \tag{1.4}$$

Compared to (1.2), an additional factor $\gamma_{Y,t^*}^{-1/2}$ appears in (1.4), which always equals 1 when $t = 1$. When $t \neq 1$, $\gamma_{Y,t^*}^{-1/2}$ is generally smaller than 1, and, unfortunately, necessary. See Remark 3.3 for some examples.

As a corollary, we have

Corollary 1.6. *Let $(L,Y)_\delta$ be a set of δ-separated lines in $\mathbb{R}^2$ with a $(\varepsilon_1,\varepsilon_2)$-two-ends shading. Let $t \in (0,2)$. Suppose that L is a Katz–Tao (δ,t)-set. Suppose also that for each $\ell \in L$, $|Y(\ell)| \geqslant \lambda\delta$, and $Y(\ell)$ is a Katz–Tao $(\delta, \min\{t,2-t\})$-set. Then for all $\varepsilon > 0$,*

$$\Big|\bigcup_{\ell\in L} Y(\ell)\Big| \geqslant c_\varepsilon \delta^\varepsilon \delta^{t\varepsilon_1/2}\lambda^{1/2}\delta^{(t-1)/2}\sum_{\ell\in L}|Y(\ell)|. \tag{1.5}$$

Notation: Throughout the paper, we use $\#E$ to denote the cardinality of a finite set. We denote by $|E|$ the Lebesgue measure of a set $E \subset \mathbb{R}^n$. If $\mathcal{E}$ is a family of sets in $\mathbb{R}^n$, we use $\cup_{\mathcal{E}}$ to denote $\cup_{E\in\mathcal{E}}E$. For $A, B \geqslant 0$, we use $A \lesssim B$ to mean $A \leqslant CB$ for an absolute (big) constant C, and use $A \sim B$ to mean $A \lesssim B$ and $B \lesssim A$. For a given $\delta < 1$, we use $A \lessapprox B$ to denote $A \leqslant c_\varepsilon\delta^{-\varepsilon}B$ for all $\varepsilon > 0$.

2. Preliminaries

We recall some basic definitions and tools in this section.

Definition 2.1. *For two finite sets E, F, we say E is a $\gtrsim$ c-refinement of F, if $E \subset F$ and $\#E \gtrsim c\#F$; we say E is a $\gtrapprox$ c-refinement of F, or simply a* **refinement** *of F, if $E \subset F$ and $\#E \gtrapprox c\#F$. A similar definition applies when E, F are two finite unions of δ-balls.*

Definition 2.2. *Let $\delta \in (0,1)$ be a small number. Let E be a finite union of δ-balls in $\mathbb{R}^n$. For another $\rho \geqslant \delta$, let*

$$|E|_\rho = \min\{\#\mathcal{D}_\rho : \mathcal{D}_\rho \text{ is a covering of } E \text{ by } \rho\text{-balls}\}. \tag{2.1}$$

Let $\mathcal{D}_\rho(E)$ be a family of ρ-balls that attains the minimum in (2.1). Denote by

$$(E)_\rho = \cup_{\mathcal{D}_\rho(E)}. \tag{2.2}$$

The same definitions apply to $\mathcal{P}$ when it is a finite set of disjoint δ-balls in $\mathbb{R}^n$ by considering $\cup_{\mathcal{P}}$.

To capture the distribution of a fractal set, we need the following two notations.

Definition 2.3 ((δ,s,C)-set)**.** *Let $\delta\in(0,1)$ be a small number. For $s\in(0,n]$, a non-empty set $E\subset\mathbb{R}^n$ is called a (δ,s,C)-**set** (or simply a (δ,s)-**set** if C is not important in the context) if*

$$|E\cap B(x,r)|_\delta\leqslant Cr^s|E|_\delta,\quad \forall x\in\mathbb{R}^n,\ r\in[\delta,1]. \tag{2.3}$$

Definition 2.4 (Katz–Tao (δ,s,C)-set)**.** *Let $\delta\in(0,1)$ be a small number. For $s\in(0,n]$, a finite set $E\subset\mathbb{R}^n$ is called a **Katz–Tao** (δ,s,C)-**set** (or simply a **Katz–Tao** (δ,s)-**set** if C is not important in the context) if*

$$\#(E\cap B(x,r))\leqslant C(r/\delta)^s,\quad \forall x\in\mathbb{R}^n,\ r\in[\delta,1]. \tag{2.4}$$

By a standard point-line duality (see, e.g., [WW24, Definition 1.13]), we have

Definition 2.5. A set of lines in $\mathbb{R}^2$ is a Katz–Tao (δ,t)-set (respectively, a (δ,t)-set) if its dual point set in $\mathbb{R}^2$ is a Katz–Tao (δ,t)-set (respectively, a (δ,t)-set).

The next two lemmas are standard. See [WW24].

Lemma 2.6. *Let $\delta\in(0,1)$ and let $C_1,C_2\geqslant 1$. If a union of δ-balls E is a (δ,s,C_1)-set and E' is a $\geqslant C_2^{-1}$-refinement of E, then E' is a (δ,s,C_1C_2)-set.*

Lemma 2.7. *Let $0<\delta<\rho<1$. Suppose E is a Katz–Tao (δ,s,C)-set, where $1\leqslant C\lessapprox 1$. Then there exists $E'\subset E$ such that $\rho^s\#E'\gtrapprox\delta^s\#E$, and E' is a Katz–Tao (ρ,s,C')-set for some $C'\lesssim 1$.*

The following is a generalization of Definition 2.3, quantifying a similar non-concentration condition, though at coarser scales.

Definition 2.8 ($(\delta,s,C;\Delta)$-set)**.** *Let $\delta\in(0,1)$ and let $\Delta\in[\delta,1]$. We say a union of δ-balls E is a $(\delta,s,C;\Delta)$-**set**, if*

$$|E\cap B(x,r)|_\delta\leqslant Cr^s|E|_\delta,\quad \forall x\in\mathbb{R}^n,\ r\in[\Delta,1]. \tag{2.5}$$

Note that a (δ,s,C)-set is a $(\delta,s,C;\Delta)$-set for all $\Delta\in[\delta,1]$. When $\Delta=\delta$, a $(\delta,s,C;\Delta)$-set is just a (δ,s,C)-set. Also, at the coarser resolution Δ, after uniformization (see Lemma 2.11), a $(\delta,s,C;\Delta)$-set is a (Δ,s,C)-set. Moreover, similar to Lemma 2.6, we have

Lemma 2.9. *Let $0<\delta\leqslant\Delta\leqslant 1$ and let $C_1,C_2\geqslant 1$. If a union of δ-balls E is a $(\delta,s,C_1;\Delta)$-set and E' is a $\gtrsim C_2^{-1}$-refinement of E, then E' is a $(\delta,s,C_1C_2;\Delta)$-set.*

Next, we recall the process of uniformization. Compared to that of an arbitrary set, the distribution of a uniform set is easier to describe. As we will see in Definition 2.12, this distribution can be captured precisely by a Lipschitz function known as the branching function of the uniform set.

Definition 2.10. *Let $\delta\in(0,1)$ be a small number. Let $M=|\log\delta|$ and let $\rho_j=M^{-j}$, $j=1,\dots,[\log_M\delta^{-1}]$. Given E a union of δ-balls in $\mathbb{R}^n$, we say E is* **uniform with error** *C if $|E\cap D_{\rho_j}|$ are the same up to a multiple of C for all $D_{\rho_j}\subset(E)_{\rho_j}$. When C is not important in the context, we say E is* **uniform**. *The same definitions apply to $\mathcal{P}$ when it is a finite set of disjoint δ-balls in $\mathbb{R}^n$ by considering $\cup_{\mathcal{P}}$.*

The following lemma shows that for an arbitrary set, we can always find a refinement that is also uniform. See [WW24, Lemma 1.11] for details.

Lemma 2.11 (Uniformization)**.** *Let $\delta \in (0,1)$ be a small number. If $E \subset \mathbb{R}^n$ is a union of dyadic δ-balls, then there is a $\gtrsim (\log|\log\delta|)^{-\frac{|\log\delta|}{\log|\log\delta|}}$-refinement E' of E such that E' is uniform.*

Definition 2.12 (Branching function)**.** *Let $\delta \in (0,1)$ and let $M = |\log\delta|$. Let $\rho_j = M^{-j}$, $j = 1, \dots, \lceil \log_M \delta^{-1} \rceil =: N$. Let $E \subset \mathbb{R}^n$ be a union of δ-balls that is uniform. Define a* **branching function** *$\beta_E : [0,N] \to [0,n]$ as*

$$\beta_E(j) = \frac{\log(|E|_{\rho_j})}{|\log\delta|}, \quad 0 \leqslant j \leqslant N \tag{2.6}$$

and interpolate linearly between $\beta_E(j)$ and $\beta_E(j+1)$.

Given a family of uniform sets, the next lemma shows that this family has a refinement whose sets have branching functions that are essentially the same. For its proof, we refer to [WW24].

Lemma 2.13. *Let $\delta \in (0,1)$, $M = |\log\delta|$, and $N = \lceil \log_M \delta^{-1} \rceil$. Suppose $\mathcal{E}$ is a finite family of uniform sets of δ-balls in the unit ball. Then there is subset $\mathcal{E}' \subset \mathcal{E}$ with $\#\mathcal{E}' \gtrsim (n \log M)^{-N} \#\mathcal{E}$ and a* **uniform branching function** *$\beta_{\mathcal{E}'}$ of $\mathcal{E}'$ such that for any $E \in \mathcal{E}'$, $|\beta_E - \beta_{\mathcal{E}'}| \leqslant |\log\delta|^{-1}$. In particular, $\#\mathcal{E}' \gtrapprox \#\mathcal{E}$.*

After rescaling, the branching function of a uniform set becomes a 1-Lipschitz function. To study such functions, Shmerkin [Shm23] introduced a powerful tool known as multi-scale decomposition, which has since been developed in various forms. The version we need is the following, whose proof can be found in [DW24].

Lemma 2.14. *Let $\eta > 0$ be a small number and let $\eta_0 = \eta_0(\eta) = \eta^{2\eta^{-1}}$. Then for a non-decreasing 1-Lipschitz function $f : [0,1] \to [0,1]$, there exists a partition*

$$0 = A_1 < A_2 < \cdots < A_{H+1} = 1 \tag{2.7}$$

and a sequence

$$0 \leqslant s_1 < s_2 < \cdots < s_H \leqslant 1 \tag{2.8}$$

such that for each $1 \leqslant h \leqslant H$, we have the following:

$$A_{h+1} - A_h \geqslant \eta_0 \eta^{-1}; \tag{2.9}$$

$$f(x) \geqslant f(A_h) + s_h(x - A_h) - \eta(A_{h+1} - A_h), \text{ for all } x \in [A_h, A_{h+1}], \tag{2.10}$$

$$f(A_{h+1}) \leqslant f(A_h) + (s_h + 3\eta)(A_{h+1} - A_h); \tag{2.11}$$

$$s_H \geqslant f(1) - f(0) - \eta. \tag{2.12}$$

What follows are several definitions related to shadings.

Definition 2.15 ($E_{L,Y}$)**.** *Let $\delta \in (0,1)$ and let $(L,Y)_\delta$ be a set of lines and shading. Define $E_{L,Y} := \bigcup_{\ell \in L} Y(\ell)$, which can be identified as a union of δ-balls. If the shading Y is apparent from the context, we will use E_L to denote $E_{L,Y}$. Moreover, for each $x \in E_{L,Y}$, define*

$$L_Y(x) = \{\ell \in L : x \in Y(\ell)\}. \tag{2.13}$$

If the shading Y is apparent from the context, we use $L(x)$ to denote $L_Y(x)$.

Definition 2.16. *Let $\delta \in (0,1)$ and let $(L,Y)_\delta$ be a set of lines and shading. We say Y is* **λ-dense**, *if $|Y(\ell)| \geqslant \lambda |N_\delta(\ell)|$.*

Definition 2.17. *Let $\delta \in (0,1)$ and let $(L,Y)_\delta$ be a family of lines and shadings. We say $(L',Y')_\delta$ is a* **refinement** *of $(L,Y)_\delta$, if $L' \subset L$, $Y'(\ell) \subset Y(\ell)$ for all $\ell \in L'$, and if*

$$\sum_{\ell' \in L'} |Y'(\ell')| \gtrapprox \sum_{\ell \in L} |Y(\ell)|. \tag{2.14}$$

Similarly, for a family of δ-tubes and shadings $(\mathcal{T},Y)$, we say $(\mathcal{T}',Y')$ is a **refinement** *of $(\mathcal{T},Y)$, if $\mathcal{T}' \subset \mathcal{T}$, $Y'(T) \subset Y(T)$ for all $T \in \mathcal{T}'$, and if $\sum_{T' \in \mathcal{T}'} |Y'(T')| \gtrapprox \sum_{T \in \mathcal{T}} |Y(T)|$.*

Although a shading $Y(\ell)$ is a union of 2-dimensional balls, it is indeed a one-dimensional object. Thus, aligning with (2.2), we make the following definition.

Definition 2.18. *Let $0 < \delta < r < 1$. Let ℓ be a line, and let $Y(\ell)$ be a shading by δ-balls. We define $(Y(\ell))_r$ as follows: Let $\mathcal{J}(\ell)$ be a minimal covering of $Y(\ell)$ by $\delta \times r$-tubes contained in $N_\delta(\ell)$. Now define $(Y(\ell))_r = \cup_{\mathcal{J}(\ell)}$.*

Definition 2.19. *Let $v, C > 0$, and let $\delta \in (0,1)$. Let ℓ be a line and $Y(\ell)$ be a uniform shading by δ-balls. Define $\rho = \rho(\ell; v, C) \in [\delta, 1]$ as*

$$\rho := \min\{r \in [\delta,1] : |Y(\ell)|_r < C^{-1} r^{-v}\}. \tag{2.15}$$

Consequently, since $Y(\ell)$ is uniform, for all $r \in [\delta, \rho]$ and all $J \subset (Y(\ell))_\rho$,

$$|Y(\ell) \cap J|_r \gtrapprox (r/\rho)^v. \tag{2.16}$$

Definition 2.19 is the standard two-ends reduction on a shading $Y(\ell)$. Note that, with $\rho = \rho(\ell; v, C)$, the ρ^{-1}-dilate of $Y(\ell) \cap J$ is a $(\delta/\rho, v, CC')$-set for all $\delta \times \rho$-tubes $J \subset (Y(\ell))_\rho$, for some $C' \lessapprox 1$. The next two lemmas were proved in [WW24].

Lemma 2.20. *Let $\delta \in (0,1)$, let ℓ be a line, and $Y(\ell)$ be a uniform shading by δ-balls. Let $0 < \varepsilon_2 < \varepsilon_1 < 1$, and let $v < \varepsilon_2$. Suppose $Y(\ell)$ is $(\varepsilon_1, \varepsilon_2, C)$-two-ends, and let $\rho = \rho(\ell; v, C)$ be the scale given by Definition 2.19. Then $\rho \geqslant \delta^{\varepsilon_1}$.*

Lemma 2.21. *Let $\delta \in (0,1)$, let ℓ be a line, and $Y(\ell)$ be a shading by δ-balls. If $Y(\ell)$ is $(\varepsilon_1, \varepsilon_2, C)$-two-ends and $Y'(\ell)$ is a refinement of $Y(\ell)$, then there exists $C' \lessapprox 1$ such that $Y'(\ell)$ is $(\varepsilon_1, \varepsilon_2, CC')$-two-ends.*

Recall Definition 1.4. The next two lemmas are about properties of γ_Y.

Lemma 2.22. *Let $0 < \delta \leqslant r \leqslant 1$ and let ℓ be a line. Suppose $Y(\ell)$ is a shading by δ-balls and $\tilde{Y}(\ell)$ is another shading by r-balls such that each r-ball in $\tilde{Y}(\ell)$ contains $\gtrsim d$ many δ-balls in $Y(\ell)$ for some $d \in [1, r/\delta]$. Then $\gamma_{\tilde{Y},t}(\ell) \lesssim (\delta/r)^{-t} d^{-1} \gamma_{Y,t}(\ell)$.*

Proof. Let ρ be the scale such that for some ball $B(x,\rho)$,

$$\gamma_{\tilde{Y},t}(\ell) \sim (r/\rho)^t |B(x,\rho) \cap \tilde{Y}(\ell)|_r. \tag{2.17}$$

Since each r-ball in $\tilde{Y}(\ell)$ contains $\sim (r/\delta)d$ many δ-balls in $Y(\ell)$, we have

$$\gamma_{Y,t}(\ell) \gtrsim (\delta/\rho)^t |B(x,\rho) \cap Y(\ell)|_r \cdot d. \tag{2.18}$$

Therefore,

$$\gamma_{\tilde{Y},t}(\ell) \lesssim (\delta/r)^{-t} d^{-1} \gamma_{Y,t}(\ell). \tag{2.19}$$

□

Lemma 2.23. *Let $(L,Y)_\delta$ be a set of lines and shading such that $\{Y(\ell):\ell\in L\}$ is a set of uniform shadings admitting a uniform branching function (recall Lemma 2.13). Let $\eta>0$ and let $\eta_0=\eta_0(\eta)=\eta^{2\eta^{-1}}$. Let $t\in[0,1]$ be given. Then there exist $r\in(\delta^{1-\eta_0\eta^{-1}},1]$ and $s\in[0,1]$ such that the following is true:*

Let $\tilde{Y}(\ell)=(Y(\ell))_r$ for all $\ell\in L$. We have

(a) $\gamma_{\tilde{Y},t}(\ell)\lesssim\gamma_{Y,t}(\ell)$.

(b) For each $\delta\times r$-tube $J\subset(Y(\ell))_r$, the r^{-1}-dilate of $Y(\ell)\cap J$ along ℓ is a $(\delta/r,s,(\delta/r)^{-9\eta})$-set, and $\log_{1/\delta}\big(\frac{|Y(\ell)|_\delta}{|Y(\ell)|_r}\big)\leqslant(s+9\eta)\log_{1/\delta}(r/\delta)$.

Proof. Let f be the uniform branching function of $\{Y(\ell):\ell\in L\}$. Apply Lemma 2.14 to f to obtain a set of scales

$$0=A_1<A_2<\cdots<A_{H+1}=1 \tag{2.20}$$

and a sequence

$$0\leqslant s_1<s_2<\cdots<s_H\leqslant 1 \tag{2.21}$$

such that the following is true

$$A_{h+1}-A_h\geqslant\eta_0\eta^{-1}; \tag{2.22}$$

$$f(x)\geqslant f(A_h)+s_h(x-A_h)-\eta(A_{h+1}-A_h),\ \text{for all } x\in[A_h,A_{h+1}], \tag{2.23}$$

$$f(A_{h+1})\leqslant f(A_h)+(s_h+3\eta)(A_{h+1}-A_h). \tag{2.24}$$

We consider two distinct cases.

Suppose $s_H\geqslant t+\eta$. Take $r=\delta^{A_H}$ and $s'_H=s_H$ so that each $\delta\times r$-segment in $(Y(\ell))_r$ contains $\gtrsim(r/\delta)^t$ many δ-balls in $Y(\ell)$. Thus, by Lemma 2.22, we have $\gamma_{\tilde{Y},t}(\ell)\leqslant\gamma_{Y,t}(\ell)$.

Suppose $s_H<t+\eta$. Take $r\geqslant\delta^{A_H}$ be the largest scale such that for all $B(x,r)\cap Y(\ell)\neq\varnothing$, we have

$$|B(x,r)\cap(Y(\ell))_{\delta^{A_H}}|_{\delta^{A_H}}\geqslant(r/\delta^{A_H})^t. \tag{2.25}$$

The choice of r implies $\gamma_{\tilde{Y},t}(\ell)=1$. Thus, item (a) follows from the simple fact that $1\leqslant\gamma_{Y,t}(\ell)$. Take s'_H be such that $\frac{|Y(\ell)|_\delta}{|Y(\ell)|_r}\sim(r/\delta)^{s'_H}$. By (2.23), we know that for all $\delta\times r$-tube $J\subset(Y(\ell))_r$,

$$|J\cap Y(\ell)|_\delta\approx(r/\delta)^{s'_H}\gtrsim(\delta^{A_H}/\delta)^{-\eta}(r/\delta^{A_H})^t(\delta^{A_H}/\delta)^{s_H}. \tag{2.26}$$

Since the sequence $\{s_j\}$ is increasing, for each $\rho\in[\delta,r]$, by (2.23) and (2.24),

$$|(Y(\ell))_\rho|_\delta\lesssim(r/\delta)^{3\eta}(\rho/\delta)^{s_H}. \tag{2.27}$$

Therefore, for all $\rho\in[\delta,r]$,

$$|J\cap Y(\ell)|_\rho\gtrapprox(r/\delta)^{-4\eta}(r/\delta^{A_H})^t(\delta^{A_H}/\rho)^{s_H}. \tag{2.28}$$

On the other hand, by (2.24), we have

$$(r/\delta)^{s'_H}\lesssim(\delta^{A_H}/\delta)^{3\eta}(r/\delta^{A_H})^t(\delta^{A_H}/\delta)^{s_H}. \tag{2.29}$$

This shows

$$|J\cap Y(\ell)|_\rho\gtrapprox(r/\delta)^{-9\eta}(r/\delta)^{s'_H}, \tag{2.30}$$

which is what we want for item (b). □

We end with two standard lemmas, whose proof can be found in [WW24].

Lemma 2.24. *Let $\delta \in (0,1)$ and let $(L,Y)_\delta$ be a set of lines in $\mathbb{R}^n$ and shading. There exists a $\mu \geqslant 1$, a set $E^\mu \subset E_L$, and a refinement $(L',Y')_\delta$ of $(L,Y)_\delta$ so that*

(1) $Y'(\ell)$ is a refinement of $Y(\ell)$ for all $\ell \in L'$.
(2) $\#L_{Y'}(x) \sim \mu$ for all $x \in E_{L,Y'}$.
(3) $Y'(\ell) = E^\mu \cap N_\delta(\ell)$ for all $\ell \in L'$.
(4) $\mu \approx |E_{L,Y'}|^{-1} \sum_{\ell \in L'} |Y(\ell')|$.

Lemma 2.25. *Let $\delta \in (0,1)$, and let $(L,Y)_\delta$ be a set of lines in $\mathbb{R}^n$ and shading. For all $x \in E_L$, there exists a $\rho = \rho(x) \in [10\delta, 1]$ so that the following is true.*

(1) There exists a refinement $L'(x) \subset L(x)$ such that $\measuredangle(\ell,\ell') + \delta \leqslant 2\rho$ for any $\ell, \ell' \in L'(x)$.
(2) There are two disjoint subsets $L_1, L_2 \subset L'(x)$ of lines such that $\#L_1, \#L_2 \gtrapprox \#L'(x)$, and $\rho \geqslant \measuredangle(\ell_1,\ell_2) \gtrapprox \rho$ for all $\ell_1 \in L_1, \ell_2 \in L_2$.

3. Proof of the main theorem

Let us restate Theorem 1.5 below.

Theorem 3.1. *Let $(L,Y)_\delta$ be a set of δ-separated lines in $\mathbb{R}^2$ with a $(\varepsilon_1,\varepsilon_2)$-two-ends, λ-dense shading. Let $t \in (0,2)$, and $t^* = \min\{t, 2-t\}$. Suppose the set of lines L is a Katz–Tao (δ,t)-set. Then for any $\varepsilon > 0$,*

$$|E_L| \geqslant c_\varepsilon \delta^\varepsilon \delta^{t\varepsilon_1/2} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y,t^*}^{-1/2} \sum_{\ell \in L} |Y(\ell)|. \tag{3.1}$$

For convenience, we introduce the following definition.

Definition 3.2. *Let $v \in (0,1]$. For a tube $T \in \mathbb{R}^n$ with cross-section radius v and length 1, let $\mathcal{L}[T]$ be the family of lines intersecting T that make an angle $\leqslant v$ with the coreline of T. Given a set of lines L, define*

$$L[T] := L \cap \mathcal{L}[T]. \tag{3.2}$$

Remark 3.3. The necessity of $\gamma_{Y,t^*}^{-1/2}$ in (3.1) can be seen by the following examples:

Suppose $r \in (\delta, 1)$. Let $(L,Y)_r$ be a set of lines and shadings appearing in [Wol99, Section 1] so that

(1) L is a Katz–Tao (r,t)-set of lines with $\#L \sim r^{-t}$.
(2) $Y(\ell)$ is a (r,s)-set of r-balls and $|Y(\ell)| \sim r^{2-s} \sim \lambda r$.
(3) $|E_L| \approx \lambda^{1/2} r^{(t-1)/2} \sum_{\ell \in L} |Y(\ell)| \sim \lambda^{3/2} r^{(1-t)/2}$.

Case 1: $t \in [0,1]$. Assume all lines in L are transversal to the horizontal axis. Let $(L',Y')_\delta$ be the anisotropic δ/r-rescaling of $(L,Y)_r$ along the horizontal direction. Then L' is a Katz–Tao (δ,t)-set of lines, and for each $\ell \in L'$, $Y'(\ell)$ is λ-dense and is a union of $\delta \times r$-segments. Clearly Y' is two-ends, and $\sum_{\ell \in L'} |Y'(\ell)| \sim \lambda r^{-t} \delta$. Moreover, we have $\gamma_{Y',t^*} \sim (r/\delta)^{1-t}$.

Now we can calculate

$$|E_{L'}| \sim (\delta/r)|E_L| \sim \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y',t^*}^{-1/2} \sum_{\ell \in L'} |Y'(\ell)|, \tag{3.3}$$

which shows the necessity of $\gamma_{Y,t^*}^{-1/2}$ in (3.1) when $t \in [0,1]$.

Case 2: $t \in [1,2]$. For each $\ell \in L$, let $L'[\ell]$ be a Katz–Tao (δ,t)-set of lines contained in $N_r(\ell)$ with $\#L'[\ell] \sim (r/\delta)^t$ and $N_r(\ell) \subset \bigcup_{\ell' \in L'[\ell]} N_\delta(\ell')$. Let $L' =$

$\bigcup_{\ell\in L} L'[\ell]$, so L' is a Katz–Tao (δ,t)-set with $\#L' \sim \delta^{-t}$. For each $\ell' \in L'[\ell]$, let $Y'(\ell') = Y(\ell) \cap N_\delta(\ell')$ be a set of $\delta \times r$-segments. Then Y' is λ-dense, and $\gamma_{Y',t*} \sim (r/\delta)^{t-1}$.

Now we can calculate

$$|E_{L'}| \sim |E_L| \sim \lambda^{1/2} r^{(t-1)/2} \sum_{\ell\in L} |Y(\ell)| \sim \lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y',t*}^{-1/2} \sum_{\ell\in L'} |Y'(\ell)|, \tag{3.4}$$

which shows the necessity of $\gamma_{Y,t*}^{-1/2}$ in (3.1) when $t \in [1,2]$.

3.1. **Outline of the proof.** For simplicity, we assume $|Y(\ell)| \sim \lambda\delta$ for all $\ell \in L$, and we only aim to heuristically justify the following simplified estimate:

$$|E_L| \gtrsim \lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t*}^{-1/2} \sum_{\ell\in L} |Y(\ell)| \tag{3.5}$$

That is, E_L has **large volume**. Note that, modulo a few steps of pigeonholing, it is equivalent to the following **small multiplicity** claim: for all $x \in E_L$,

$$\#L(x) \lesssim \lambda^{-1/2}\delta^{(1-t)/2}\gamma_{Y,t*}^{1/2}. \tag{3.6}$$

We will show that either (3.5) or (3.6) holds, which implies Theorem 3.1.

We assume that $Y(\ell)$ admits the same branching function for all $\ell \in L$. By Lemma 2.23, there is a parameter s and a scale r such that the r^{-1}-dilate of the shading $J \cap Y(\ell)$ is a $(\delta/r, s)$-set and $|J \cap Y(\ell)|_\delta \approx (\delta/r)^{-s}$ for all $\delta \times r$-tube $J \subset (Y(\ell))_r$ (see Definition 2.18).

Consider the family $\mathcal{T} = \{J \subset (Y(\ell))_r : \ell \in L\}$ of $\delta \times r$-tubes generated by $(L, Y)_\delta$. We also introduce the new shading $\bar{Y}(\ell) := (Y(\ell))_r$, which consists of a union of $\delta \times r$-tubes and is $\lambda(r/\delta)^{1-s}$-dense. Notice that, by Lemma 2.23, we have $\gamma_{\bar{Y},t*} \lesssim \gamma_{Y,t*}$. Moreover, for each $\delta/r \times 1$-tube $\bar{T}$, the anisotropic r/δ-dilate of the configuration $(L[\bar{T}], \bar{Y})_\delta$ along the normal direction of $\bar{T}$ yields a Katz–Tao (r,t)-set of lines in $\mathbb{R}^2$ with two-ends, $\lambda(r/\delta)^{1-s}$-dense shadings. Thus by induction, we can apply (3.6) at scale r to conclude that for all $\delta \times r$-tubes $J \in \mathcal{T}$

$$\#\{\ell \in L : J \subset (Y(\ell))_r\} \sim \#\{\ell \in L[\bar{T}] : J \subset (Y(\ell))_r\} \tag{3.7}$$

$$\lesssim (\lambda(r/\delta)^{1-s})^{-1/2} r^{(1-t)/2} \gamma_{\bar{Y},t*}^{1/2}. \tag{3.8}$$

Here $\bar{T}$ is the $(\delta/r) \times 1$-tube in the same direction as J.

Next, we are going to analyze what happens inside an r-ball B. Let $\mathcal{T}_B$ be the set of $\delta \times r$-tubes in $\mathcal{T}$ that intersect the r-ball B. For these $\delta \times r$-tubes we define a shading Y_B as follows. For each $J \in \mathcal{T}_B$, we choose a line $\ell \in L$ such that $J \subset (Y(\ell))_r$, and then define $Y_B(J) = J \cap Y(\ell)$. As a result, the r^{-1}-dilate of the shading $Y_B(J)$ is a $(\delta/r, s)$-set.

Suppose $\mathcal{T}_B$ is uniform, in the sense that its r^{-1}–dilate is a uniform set of $\delta/r \times 1$–tubes. By the standard broad-narrow reduction, we may assume that all points in $E_{\mathcal{T}_B,Y_B}$ are broad. That is, applying Lemma 2.25 to the input $(\mathcal{T}_B, Y_B)$ yields $\rho \gtrapprox 1$. In particular, this implies that, after point-line duality, the r^{-1}-dilate of (T_B, Y_B) obeys the assumption of Theorem 3.8. Thus, we obtain three possible outcomes (see Proposition 3.10):

(1) **(Small multiplicity):** For each typical $x \in E_{\mathcal{T}_B,Y_B}$, we have

$$\#\mathcal{T}_B(x) \lessapprox (\delta/r)^{(s-t)/2}. \tag{3.9}$$

(2) **(Large volume):** For each typical r-ball B, we have

$$|E_L \cap B| \gtrapprox (\delta/r)^{2-(t+3s)/2}|B|. \tag{3.10}$$

(3) **(Thickened tubes):** There exists a scale Δ much larger than δ such that for each typical $J \in \mathcal{T}_B$, we have

$$|E_L \cap N_\Delta(J)| \gtrapprox (\delta/r)^{1-s}|J|. \tag{3.11}$$

Finally, we show that the main theorem follows in either case.

Case I. Suppose (3.9) is true. By (3.7), we have

$$\#L(x) \lesssim (\lambda(r/\delta)^{1-s})^{-1/2} r^{(1-t)/2}\gamma_{\bar{Y},t*}^{1/2} \cdot (\delta/r)^{(s-t)/2} \lesssim \lambda^{-1/2}\delta^{(1-t)/2}\gamma_{\bar{Y},t*}^{1/2}. \tag{3.12}$$

This yields (3.6) as $\gamma_{\bar{Y},t*} \lesssim \gamma_{Y,t*}$.

Case II. Suppose (3.10) is true. Consider a new shading $\tilde{Y}(\ell) = N_r(Y(\ell))$ for a typical $\ell \in L$. By assumption, it is $\tilde{\lambda}$-dense and is two-ends as well, where

$$\tilde{\lambda} := \lambda(r/\delta)^{1-s}. \tag{3.13}$$

Now we refine L to a Katz–Tao (r,t)-set. More precisely, apply Lemma 2.7 to obtain a $\gtrapprox (\delta/r)^t$-refinement $\tilde{L}$ of L such that $\tilde{L}$ is a Katz–Tao (r,t)-set. By induction, apply (3.5) at scale r so that

$$|E_{\tilde{L},\tilde{Y}}| \gtrsim \tilde{\lambda}^{1/2} r^{(t-1)/2}\gamma_{\tilde{Y},t*}^{-1/2} \sum_{\ell\in\tilde{L}} |\tilde{Y}(\ell)| \tag{3.14}$$

$$\gtrsim (\tilde{\lambda}/\lambda)^{3/2}\lambda^{1/2} r^{(t-1)/2}\gamma_{\tilde{Y},t*}^{-1/2}(\delta/r)^{t-1} \sum_{\ell\in L} |Y(\ell)|. \tag{3.15}$$

Finally, combining (3.10), (3.14), and (3.13), and noting $\gamma_{\tilde{Y},t*} \lesssim \gamma_{Y,t*}$, we have

$$|E_L| \sim \min_{B\in\mathcal{B}_{r,6}} \frac{|E_L \cap B|}{|B|} \cdot |E_{\tilde{L},\tilde{Y}}| \tag{3.16}$$

$$\gtrapprox (\delta/r)^{2-(t+3s)/2} \cdot (\tilde{\lambda}/\lambda)^{3/2}\lambda^{1/2} r^{(t-1)/2}\gamma_{\tilde{Y},t*}^{-1/2}(\delta/r)^{t-1} \sum_{\ell\in L} |Y(\ell)| \tag{3.17}$$

$$\gtrsim (\delta/r)^{2-(t+3s)/2}(\tilde{\lambda}/\lambda)^{3/2}(\delta/r)^{(t-1)/2} \cdot \lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t*}^{-1/2} \sum_{\ell\in L} |Y(\ell)| \tag{3.18}$$

$$\gtrsim \lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t*}^{-1/2} \sum_{\ell\in L} |Y(\ell)|. \tag{3.19}$$

Case III. Suppose (3.11) is true. Consider the thickened shading $\tilde{Y}(\ell) = E_L \cap N_\Delta(\ell)$. Then (3.11) implies that $\tilde{Y}(\ell)$ is still $\gtrapprox \lambda$-dense for each typical $\ell \in L$. That is, passing to the thickened configuration $(L,\tilde{Y})_\Delta$ incurs essentially no loss. Thus, we may apply (3.5) at scale Δ to conclude the theorem. We remark that some additional technical work is required to control the Katz–Tao constant $\gamma_{\tilde{Y},t*}$ for this thickened shading; we defer these details to the formal proof.

3.2. **Some incidence results and two-ends reduction.** The proof of Theorem 3.1 relies on several incidence estimates.

Lemma 3.4. *Let $0 < t < u \leqslant 1$ and let $\eta \in (0,(u-t)/2)$. Let $\mathcal{B}$ be a collection of δ-balls in $\mathbb{R}^2$, and for each $B \in \mathcal{B}$, let $\mathcal{T}(B)$ be a family of $\delta\times 1$-tubes intersecting B. Suppose $\cup_{\mathcal{B}}$ is a $(\delta, 2-t, \delta^{-\eta})$-set. Suppose also that for each $B \in \mathcal{B}$, $\mathcal{T}(B)$,*

identified with a set of delta-arcs, is a $(\delta, u, \delta^{-\eta})$-set. Let r be such that $\#\mathcal{T}(B) \geqslant r$ for all $B \in \mathcal{B}$. Then for any $\varepsilon > 0$,

$$\#\bigcup_{B\in\mathcal{B}} \mathcal{T}(B) \geqslant c_\varepsilon \delta^\varepsilon \delta^{-1} r, \tag{3.20}$$

where $\#\bigcup_{B\in\mathcal{B}} \mathcal{T}(B)$ should be understood as the maximum number of distinct tubes contained in $\bigcup_{B\in\mathcal{B}} \mathcal{T}(B)$ (two $\delta\times 1$-tubes T_1 and T_2 are distinct if $|T_1\cap T_2| \leqslant |T_1|/2$).

Lemma 3.4 was stated in [RW23, Theorem 4.1] with an additional dependence between η and ε. However, this dependence can be removed since we are only considering the upper range in the Furstenberg set estimate. We refer to [RW23, Proposition 4.2] for its proof.

Lemma 3.5. *Let $0 < t < u \leqslant 1$ and let $\eta \in (0, (u-t)/4)$. Let $\Delta \in [\delta^{1-\eta}, 1]$, and let $\mathcal{B}$ be a set of δ-balls in $\mathbb{R}^2$ contained in some Δ-ball B_Δ. For each $B \in \mathcal{B}$, let $\mathcal{T}(B)$ be a set of $\delta \times 1$-tubes intersecting B. Suppose that the Δ^{-1}-dilate of $\mathcal{B}$ is a $((\delta/\Delta), 2-t, (\delta/\Delta)^{-\eta})$-set, and for each $B \in \mathcal{B}$, $\mathcal{T}(B)$ is a $(\delta, u, (\delta/\Delta)^{-\eta})$-set. Let $r \geqslant 1$ be such that $\#\mathcal{T}(B) \geqslant r$ for all $B \in \mathcal{B}$. Then for any $\varepsilon > 0$,*

$$\#\bigcup_{B\in\mathcal{B}} \mathcal{T}(B) \geqslant c_\varepsilon \delta^\varepsilon (\Delta/\delta) r. \tag{3.21}$$

Proof. For each $B \in \mathcal{B}$, let $\mathcal{T}_{\delta/\Delta}(B)$ be the family of distinct $\delta/\Delta\times 1$-tubes containing at least one $\delta \times 1$-tube in $\mathcal{T}(B)$. For each $T_{\delta/\Delta} \in \mathcal{T}_{\delta/\Delta}(B)$, define $\mathcal{T}(T_{\delta/\Delta}) := \{T \in \mathcal{T}(B) : T \subset T_{\delta/\Delta}\}$. By dyadic pigeonholing on $\{\#\mathcal{T}(T_{\delta/\Delta}) : T_{\delta/\Delta} \in \mathcal{T}_{\delta/\Delta}(B)\}$, there exists a set $\mathcal{T}'_{\delta/\Delta}(B) \subset \mathcal{T}_{\delta/\Delta}(B)$ and a number μ_B such that $\#\mathcal{T}(T_{\delta/\Delta}) \sim \mu_B$ for all $T_{\delta/\Delta} \in \mathcal{T}'_{\delta/\Delta}(B)$ and $\#\mathcal{T}'_{\delta/\Delta}(B) \cdot \mu_B \gtrapprox \#\mathcal{T}(B) \geqslant r$. Since $\mathcal{T}(B)$ is a $(\delta, u, (\delta/\Delta)^{-\eta})$-set, $\mathcal{T}_{\delta/\Delta}(B)$ is a $(\delta/\Delta, u, (\delta/\Delta)^{-\eta})$-set, yielding that $\mathcal{T}'_{\delta/\Delta}(B)$ is a $(\delta/\Delta, u, (\delta/\Delta)^{-2\eta})$-set.

For each $\delta/\Delta \times 1$-tube $T_{\delta/\Delta} \in \mathcal{T}'_{\delta/\Delta}(B)$, there is a unique $\delta \times \Delta$-tube $\bar{T}$ intersecting B that is parallel to $T_{\delta/\Delta}$. Let $\bar{\mathcal{T}}'(B)$ be this set of $\delta \times \Delta$-tubes, so $\#\bar{\mathcal{T}}'(B) = \#\mathcal{T}'_{\delta/\Delta}(B) \gtrapprox r/\mu_B$. Note that $\bar{\mathcal{T}}'(B)$ is indeed the Δ-dilate of $\mathcal{T}'_{\delta/\Delta}(B)$. In particular, the Δ^{-1}-dilate of $\bar{\mathcal{T}}'(B)$ is a $(\delta/\Delta, u, (\delta/\Delta)^{-2\eta})$-set. By dyadic pigeonholing on $\{\mu_B : B \in \mathcal{B}\}$, there exists a refinement $\mathcal{B}'$ of $\mathcal{B}$ and a uniform μ such that $\mu_B \sim \mu$ for all $B \in \mathcal{B}'$. Hence, the Δ^{-1}-dilate of $\mathcal{B}'$ is a $((\delta/\Delta), 2-t, (\delta/\Delta)^{-2\eta})$-set. Moreover, $\#\mathcal{T}(T_{\delta/\Delta}) \sim \mu$ for all $T_{\delta/\Delta} \in \mathcal{T}'_{\delta/\Delta}(B)$ and all $B \in \mathcal{B}'$. This shows that there are $\gtrsim \mu$ many $\delta \times 1$-tubes in $\mathcal{T}(B)$ intersecting $\bar{T}$ for all $\bar{T} \in \bar{\mathcal{T}}'(B)$ and all $B \in \mathcal{B}'$. Therefore, $\#\bigcup_{B\in\mathcal{B}} \mathcal{T}(B) \gtrsim \mu \#\bigcup_{B\in\mathcal{B}'} \bar{\mathcal{T}}'(B)$.

Apply a Δ-dilate version of Lemma 3.4 to the $\delta \times \Delta$-tubes $\{\bar{\mathcal{T}}'_B, B \in \mathcal{B}'\}$ so that

$$\#\bigcup_{B\in\mathcal{B}} \mathcal{T}(B) \gtrsim \mu \cdot \#\bigcup_{B\in\mathcal{B}'} \bar{\mathcal{T}}'(B) \gtrapprox \mu(\Delta/\delta)(r/\mu) \gtrapprox (\Delta/\delta) r. \tag{3.22}$$

□

To prove the next result, we need the following Furstenberg set estimate proved in [OS23] and [RW23].

Theorem 3.6 ([RW23], Theorem 4.1, Remark 6.8)**.** *For all $\nu > 0$, there exists an $\eta > 0$ such that the following is true for all $s \in [0, 1]$ and $t \in [0, 2]$.*

Let $(L, Y)_\delta$ be a set of lines and shadings such that L is a $(\delta, t, \delta^{-\eta})$-set, and $Y(\ell)$ is a $(\delta, s, \delta^{-\eta})$-set for all $\ell \in L$. Then

$$|E_L| \gtrsim \delta^\nu \delta^2 \delta^{-\min\{t+s, \frac{3s+t}{2}, 1+s\}}. \tag{3.23}$$

Lemma 3.7. *Let (ν,η) be as in Theorem 3.6. Let $\lambda\in(0,1]$. Let $(L,Y)_\delta$ be a set of lines and λ-dense shadings such that L is a Katz–Tao (δ,t)-set. Suppose for all $\ell\in L$, $Y(\ell)$ is a $(\delta,s,\delta^{-\eta})$-set, and $|Y(\ell)|\lesssim\delta^{1-s-\eta}|N_\delta(\ell)|$. Then*

$$|E_L|\gtrsim\delta^{\nu+O(\eta)}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t*}^{-1/2}\sum_{\ell\in L}|Y(\ell)|. \tag{3.24}$$

Proof. By a standard probabilistic argument (see [DW24, Section 3]), it suffices to prove (3.24) with the additional assumption that L is a (δ,t)-set.

In this case, apply Theorem 3.6 to have

$$|E_L|\gtrsim\delta^{\nu}\delta^{2}\delta^{-\min\{t+s,\frac{3s+t}{2},1+s\}}\gtrsim\delta^{\nu+O(\eta)}\delta^{\max\{0,\frac{t-s}{2},t-1\}}\sum_{\ell\in L}|Y(\ell)| \tag{3.25}$$

$$\gtrsim\delta^{\nu+O(\eta)}\delta^{\max\{\frac{s-t}{2},0,\frac{t+s}{2}-1\}}\lambda^{1/2}\delta^{(t-1)/2}\sum_{\ell\in L}|Y(\ell)|. \tag{3.26}$$

To prove (3.24), it suffices to show $\gamma_{Y,t*}\gtrsim\delta^{O(\eta)}\delta^{\min\{t-s,\,0,\,2-t-s\}}$. But this is obviously true, as $|Y(\ell)|\gtrsim\delta^{1-s+\eta}|N_\delta(\ell)|$. □

The next theorem was proved in [DW24]. We state a more quantitative version of it here.

Theorem 3.8 ([DW24], Theorem 5.4)**.** *For any $\nu>0$, there exists an $\eta>0$, which is much smaller than $\nu^{2\nu^{-1}}$, such that the following is true for sufficiently small δ:*

Let $\mathcal{T}$ be a family of $\delta\times1$-tubes in the plane, and let $\mathcal{P}$ be a family of uniform δ-balls such that for any $\rho\in[\delta,1]$,

$$|\cup_{\mathcal{P}}|_\rho\gtrsim\rho^{-s}\delta^{\eta}. \tag{3.27}$$

For each δ-ball $p\in\mathcal{P}$, let $\mathcal{T}_p\subset\mathcal{T}$ be a $(\delta,s,\delta^{-\eta})$-set (here we identify each $T\in\mathcal{T}_p$ as a δ-arc contained in $\mathbb{S}^1$) of $\delta\times1$-tubes passing through p with $\#\mathcal{T}_p\sim r$. Moreover, as a union of δ-arcs of $\mathbb{S}^1$, $\cup_{\mathcal{T}_p}$ is uniform, and there is a uniform branching function of the family of the union of δ-arcs $\{\cup_{\mathcal{T}_p}:p\in\mathcal{P}\}$.

Then one of the following must be true:

(1) We have

$$\#\mathcal{P}\lesssim\delta^{s-\nu}\frac{(\#\mathcal{T})^2}{r^2}. \tag{3.28}$$

(2) There exists a scale $\Delta\gtrsim\delta^{1-\sqrt{\eta}}$ such that for each Δ-ball $B\subset(\cup_{\mathcal{P}})_\Delta$, the Δ^{-1}-dilate of $\cup_{\mathcal{P}}\cap B$ is a uniform, $(\delta/\Delta,2-s+\eta^{1/4},\delta^{-\eta})$-set.

Remark 3.9. In [DW24, Theorem 5.4], Item (2) is stated in a weaker way. However, their proof indeed gives the current stronger version of Item (2), as $\mathcal{P}$ is uniform.

We will use Theorem 3.8 to prove the following proposition, which is a combination of Lemma 3.5 and the dual version of Theorem 3.8.

Proposition 3.10. *For any $\nu>0$, there exists an $\eta>0$, which is much smaller than $\nu^{2\nu^{-1}}$, such that the following is true for sufficiently small δ:*

Let $(L,Y)_\delta$ be a set of uniform, δ-separated lines such that for all $\ell\in L$ we have

(1) $Y(\ell)$ is a $(\delta,s,\delta^{-\eta/2})$-set, yielding $|Y(\ell)|\gtrsim\delta^{1-s+\eta/2}|N_\delta(\ell)|$.

(2) $|Y(\ell)|\lesssim\delta^{1-s-\eta/2}|N_\delta(\ell)|$.

Suppose in addition that

(i) For each $\ell \in L$, the shading $Y(\ell)$ is uniform, and there is a uniform branching function for the family of shadings $\{Y(\ell) : \ell \in L\}$.
(ii) For all $\rho \in [\delta, 1]$, there are $\gtrsim \rho^{-s}\delta^{\eta/2}$ many distinct $\rho \times 1$-tubes T_ρ such that $L[T_\rho] \neq \varnothing$.
(iii) $\#L(x) \lessapprox |E_L|^{-1} \sum_{\ell \in L} |Y(\ell)|$ for any $x \in E_L$.

Then there exists a refinement $(L', Y')_\delta$ of $(L, Y)_\delta$ such that one of the following must be true for all $t \in [0, 2]$:

(1) $\#L'(x) \lessapprox \delta^{(s-t)/2}$ for all $x \in E_{L'}$.
(2) We have

$$|E_{L'}| \gtrsim \delta^{\nu+3\eta}\delta^{2-(t+3s)/2}. \tag{3.29}$$

(3) There is a $\Delta \gtrsim \delta^{1-\sqrt{\eta}}$, and for each $\ell \in L'$, two numbers $d_1(\ell), d_2(\ell)$ and two families of disjoint Δ-balls $\mathcal{Q}_1(\ell), \mathcal{Q}_2(\ell)$ such that the following is true:
(a) $\#\mathcal{Q}_j(\ell) = d_j(\ell), j = 1, 2$, and $\cup_{\mathcal{Q}_2(\ell)} \subset \cup_{\mathcal{Q}_1(\ell)} \subset N_\Delta(\ell)$.
(b) $|E_L \cap Q| \gtrsim \delta^{1-s+2\eta} d_1(\ell)^{-1}\Delta^{-1}|Q|$ for each $Q \in \mathcal{Q}_1(\ell)$.
(c) For each $Q \in \mathcal{Q}_2(\ell)$, Q contains $\gtrsim \delta^{-s-\eta} d_2(\ell)^{-1}$ many δ-balls in $Y'(\ell)$.

Remark 3.11. The factor $d_2(\ell)$ appearing in Case (3) is not important to the numerological aspect of the proof of Theorem 3.1. It was introduced to indicate that $Y'(\ell)$ is essentially contained in the set $\cup_{\mathcal{Q}_1(\ell)}$. The reader may take $d_1(\ell) = d_2(\ell)$ and $\mathcal{Q}_1(\ell) = \mathcal{Q}_2(\ell)$ for convenience.

Proof of Proposition 3.10. Apply Lemma 2.24 to $(L, Y)_\delta$ to obtain a refinement (L, Y_1) of $(L, Y)_\delta$ so that $\#L_{Y_1}(x)$ is the same up to a constant multiple for all $x \in E_{L,Y_1}$. Apply Lemma 2.11 to E_{L,Y_1} to obtain a uniform refinement E_1 of E_{L,Y_1}. Define $Y_2(\ell) = Y_1(\ell) \cap E_1$, so that $(L, Y_2)_\delta$ is a refinement of $(L, Y)_\delta$. Therefore, by dyadic pigeonholing, we can find a refinement $(L_2, Y_2)_\delta$ of $(L, Y_2)_\delta$ such that L_2 is uniform and

(1) For all $\ell \in L_2$, $Y_2(\ell)$ is uniform and is a refinement of $Y(\ell)$. As a result, $Y_2(\ell)$ is a $(\delta, s, \delta^{-\eta})$-set and $|Y_2(\ell)| \lessapprox \delta^{1-s-\eta}|N_\delta(\ell)|$
(2) For each $\ell \in L_2$, the shading $Y_2(\ell)$ is uniform, and there is a uniform branching function for the family of shadings $\{Y_2(\ell) : \ell \in L_2\}$.
(3) For all $\rho \in [\delta, 1]$, there are $\gtrsim \rho^{-s}\delta^{\eta}$ many distinct $\rho \times 1$-tubes T_ρ such that $L_2[T_\rho] \neq \varnothing$.

By the point-line duality, we can identify L_2 as a set of points $\mathcal{P} \subset [0, 1]^2$ with the following correspondences:

(1) Each $\ell \in L_2$ corresponds to a point $p \in \mathcal{P}$, and the shading $Y_2(\ell)$ corresponds to a family of $\delta \times 1$-tubes $\mathcal{T}_p$ passing through a point $p \in \mathcal{P}$.
(2) The union of δ-balls E_{L_2} corresponds to the union of δ-tubes $\mathcal{T} = \cup_{p\in\mathcal{P}}\mathcal{T}_p$.

It is straightforward to check that the configuration $(\mathcal{P}, \cup_{p\in\mathcal{P}}\mathcal{T}_p)$ obeys the hypothesis of Theorem 3.8, with $r \gtrsim \delta^{-s+\eta}$. Thus, one of the following must be true:

A. We have

$$\#\mathcal{P} \lesssim \delta^{s-\nu}\frac{(\#\mathcal{T})^2}{r^2}. \tag{3.30}$$

B. There exists a scale $\Delta \gtrsim \delta^{1-\sqrt{\eta}}$ such that for each Δ-ball $B \subset (\cup_{\mathcal{P}})_\Delta$, the Δ^{-1}-dilate of $\cup_{\mathcal{P}} \cap B$ is a uniform, $(\delta/\Delta, 2 - s + \eta^{1/4}, \delta^{-\eta})$-set.

Suppose Case **A** happens. Recall that $r \gtrsim \delta^{-s+\eta}$. By reversing point-line duality, we have

$$|E_{L_2}| \gtrsim \delta^{\nu/2+\eta}\delta^{2-3s/2}(\#L_2)^{1/2}. \tag{3.31}$$

Recall that $\#L(x) \lessapprox |E_L|^{-1}\sum_{\ell\in L}|Y(\ell)|$ for all $x \in E_L$ and $|Y(\ell)| \lesssim \delta^{1-s-\eta}|N_\delta(\ell)|$ for any $\ell \in L$. Since $(L_2, Y_2)_\delta$ is a refinement of $(L, Y)_\delta$, we have $\#L_2(x) \lessapprox |E_{L_2}|^{-1}\delta^{2-s-\eta}(\#L_2) \lessapprox \delta^{-\nu/2-2\eta}\delta^{s/2}(\#L_2)^{1/2}$.

Take $(L', Y')_\delta = (L_2, Y_2)_\delta$. If $\#L \leqslant \delta^{-t+3\eta+\nu}$, then $\#L_2(x) \lessapprox \delta^{(s-t)/2}$ for any $x \in E_{L_2}$, giving item (1) in Proposition 3.10. If $\#L \geqslant \delta^{-t+3\eta+\nu}$, then $|E_{L_2}| \gtrsim \delta^{\nu+3\eta}\delta^{2-(t+3s)/2}$, giving item (2) in Proposition 3.10.

Suppose Case **B** happens. Then L_2 is the set of lines L' we needed in the Proposition. As for the rest, consider each Δ-ball $B \subset (\cup\mathcal{P})_\Delta$ and let $\mathcal{P}_B$ be the set of δ-balls in $\mathcal{P}$ contained in B. For each $p \in \mathcal{P}_B$, consider the family of tubes $\mathcal{T}_p$, which is a $(\delta, s, \delta^{-\eta})$-set. Since $\Delta \gtrsim \delta^{1-\sqrt{\eta}}$, $\mathcal{T}_p$ is a $(\delta, s, (\delta/\Delta)^{-\sqrt{\eta}})$-set for all $p \in \mathcal{P}_B$. Similarly, since the Δ^{-1}-dilate of the $\mathcal{P}_B$ is a $(\delta/\Delta, 2-s+\eta^{1/4}, \delta^{-\eta})$-set, it is also a $(\delta/\Delta, 2-s+\eta^{1/4}, (\delta/\Delta)^{-\sqrt{\eta}})$-set. Applying Lemma 3.5 to the configuration $(\mathcal{P}_B, \{\mathcal{T}_p\}_{p\in\mathcal{P}_B})$ with $(u, t, \eta, r) = (s, s-\eta^{1/4}, \eta^{1/2}, \delta^{-s+\eta})$, we have $\#\bigcup_{p\in\mathcal{P}_B}\mathcal{T}_p \gtrapprox (\Delta/\delta)\delta^{-s+\eta}$.

By reversing the point-line duality, each Δ-ball B corresponds to a Δ-tube T_Δ, and the $\delta\times 1$-tubes $\bigcup_{p\in\mathcal{P}_B}\mathcal{T}_p$ correspond to the δ-balls $\bigcup_{\ell\subset T_\Delta} Y_2(\ell)$. Thus, what we had above can be stated as follows: There is a partition of L_2 into $\{L_2[T_\Delta], T_\Delta \in \mathcal{T}_\Delta\}$, where $\mathcal{T}_\Delta$ is a family of Δ-separated $\Delta\times 1$-tubes, and $L_2[T_\Delta]$ is the set of $\ell \in L_2$ contained in T_Δ, such that

$$\Big|\bigcup_{\ell\in L_2[T_\Delta]} Y_2(\ell)\Big| \sim \delta^2 \#\bigcup_{p\in\mathcal{P}_B}\mathcal{T}_p \gtrapprox (\Delta/\delta)\delta^{2-s+\eta}. \tag{3.32}$$

Since E_1 is uniform, there exists a number α such that $|E_1\cap Q| \sim \alpha|Q|$ for all $Q \in \mathcal{D}_\Delta(E_1)$. For each $\ell \in L_2$, let $\mathcal{Q}_1(\ell)$ be the family of disjoint Δ-balls contained in $N_\Delta(\ell)$ such that $E_1 \cap Q \neq \varnothing$ for all $Q \in \mathcal{Q}_1(\ell)$. Let $T_\Delta(\ell)$ be the Δ-tube that contains ℓ. Since $\bigcup_{\ell'\in L_2[T_\Delta(\ell)]} Y_2(\ell') \subset E_1 \cap N_\Delta(\ell)$, by (3.32), we have

$$\#\mathcal{Q}_1(\ell)\alpha\Delta^2 \sim \sum_{Q\in\mathcal{Q}_1(\ell)} |E_1\cap Q| \sim |E_1 \cap \cup_{\mathcal{Q}_1(\ell)}| \gtrapprox (\Delta/\delta)\delta^{2-s+\eta}. \tag{3.33}$$

This gives item (b) in Case (3).

For items $(a), (c)$ in Case (3), let $\mathcal{Q}_2(\ell) \subset \mathcal{Q}_1(\ell)$ be the family of Δ-balls such that $Q \cap Y_2(\ell) \neq \varnothing$ for all $Q \in \mathcal{Q}_2(\ell)$. Since $Y_2(\ell)$ is uniform and since $Y_2(\ell) \subset E_1$, we know that $\mathcal{Q}_2(\ell) \subset \mathcal{Q}_1(\ell)$. Finally, since $Y_2(\ell)$ is a uniform $(\delta, s, \delta^{-\eta})$-set, Q contains $\gtrsim \delta^{-s-\eta}d_2(\ell)^{-1}$ many δ-balls in $Y_2(\ell)$. □

Similar to the argument in [WW24], Theorem 3.1 reduces to the following proposition via a standard two-ends reduction.

Proposition 3.12. *Let $\varepsilon > 0$ be given. Let $\eta > 0$ be such that Proposition 3.10 and Lemma 3.7 are true with $(\nu, \eta) = (\varepsilon^2, 2\eta)$. Then there exists $c_\varepsilon > 0$ such that the following is true for all $\delta \in (0, 1)$:*

Let $(L, Y)_\delta$ be a set of δ-separated lines in $\mathbb{R}^2$ with a uniform, λ-dense shading. Let $t \in (0, 2)$ and $t^* = \min\{t, 2-t\}$. Suppose L is a Katz–Tao (δ, t)-set, and suppose

$Y(\ell)$ is a $(\delta, \varepsilon^2, C; \rho)$-set for some $\rho \in [\delta, \delta^\eta]$ and for all $\ell \in L$. Then

$$|E_L| \geqslant c_\varepsilon \delta^\varepsilon C^{-\eta^{-2}} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y,t*}^{-1/2} \sum_{\ell \in L} |Y(\ell)|. \tag{3.34}$$

Proof of Theorem 3.1 via Proposition 3.12. It suffices to prove Theorem 3.1 for all $\varepsilon \in (0, \sqrt{\varepsilon_2})$. By dyadic pigeonholing, there exists a refinement L_1 of L such that for all $\ell \in L_1$, $|Y(\ell)|$ are the same up to a constant multiple. Let $Y_1 = Y$, and assume $|Y_1(\ell)| \sim \lambda\delta$ for all $\ell \in L_1$, without loss of generality.

Apply Lemma 2.11 and Lemma 2.13 to the set of shadings $\{Y_1(\ell) : \ell \in L_1\}$, we know that there is a refinement $(L_2, Y_2)_\delta$ of $(L_1, Y_1)_\delta$ such that

(1) For all $\ell \in L_2$, $Y_2(\ell)$ is uniform, and $Y_2(\ell)$ is a refinement of $Y_1(\ell)$.
(2) There is a uniform branching function β_{L_2} for the shadings $\{Y_2(\ell) : \ell \in L_2\}$.

By Lemma 2.21, there exists $K \lessapprox 1$ such that Y_2 is $(\varepsilon_1, \varepsilon_2, K)$-two-ends. For each $\ell \in L_2$, by Definition 2.19, there exists a scale $\rho(\ell) = \rho(\ell; \varepsilon^2, K) \in [\delta, 1]$ such that

(1) $|Y_2(\ell)|_{\rho(\ell)} < K^{-1}\rho(\ell)^{-\varepsilon^2}$.
(2) For all $r \in [\delta, \rho(\ell)]$ and all $J \subset (Y_2(\ell))_{\rho(\ell)}$, $|Y_2(\ell) \cap J|_r \gtrapprox K^{-1}(r/\rho)^{\varepsilon^2}$.

Since $\varepsilon^2 < \varepsilon_2$, by Lemma 2.20, $\rho(\ell) \geqslant \delta^{\varepsilon_1}$ for all $\ell \in L_2$. Apply dyadic pigeonholing to the set $\{\rho(\ell) : \ell \in L_2\}$, we can find a uniform ρ and a refinement L_3 of L_2 such that $\rho(\ell) = \rho$ for all $\ell \in L_3$. Let $Y_3 = Y_2$. Then the set of lines and shading $(L_3, Y_3)_\delta$ is a refinement of $(L, Y)_\delta$ with a uniform, λ-dense shading. Moreover, for all $\ell \in L_3$, the following is true:

(1) $|Y_3(\ell)|_\rho < K^{-1}\rho^{-\varepsilon^2} \lessapprox \rho^{-\varepsilon^2}$.
(2) For all $r \in [\delta, \rho]$ and all $J \subset (Y_3(\ell))_\rho$, $|Y_3(\ell) \cap J|_r \gtrapprox K^{-1}(r/\rho)^{\varepsilon^2}$.
(3) $\rho \geqslant \delta^{\varepsilon_1}$.

Let $\mathcal{B}_\rho$ be a family of finite-overlapping ρ-balls that covers E_{L_3}. For each $B \in \mathcal{B}_\rho$, let $\mathcal{T}_B^0$ be the family of distinct $\delta \times \rho$-tubes in B. Define $\mathcal{T}_B$ as

$$\mathcal{T}_B = \{J \in \mathcal{T}_B^0 : J \subset (Y_3(\ell))_\rho \text{ for some } \ell \in L_3\}. \tag{3.35}$$

For each $J \in \mathcal{T}_B$, let $L_3(J) = \{\ell \in L_3 : J \subset (Y_3(\ell))_\rho\}$. Pick one $\ell \in L_3(J)$ to define a shading Y_B on J as

$$Y_B(J) = Y_3(\ell) \cap J. \tag{3.36}$$

Since Y_3 is uniform, we know that $Y_B(J)$ is uniform, and $|Y_B(J)|$ are the same up to a constant multiple for all $J \in \mathcal{T}_B$. Since Y_3 is $\gtrapprox \lambda$-dense and since $|Y_3(\ell)|_\rho \lessapprox \rho^{-\varepsilon^2}$, $Y_B(J)$ is $\gtrapprox \lambda\rho^{-1+\varepsilon^2}$-dense. Recall that for all $r \in [\delta, \rho]$ and all $T \subset (Y_3(\ell))_\rho$, $|Y_3(\ell) \cap T|_r \gtrapprox K^{-1}(r/\rho)^{\varepsilon^2}$. Thus, the ρ^{-1}-dilate of $Y_B(J)$ is a $(\delta/\rho, \varepsilon^2, K')$-set for some $K' \lessapprox 1$.

We want to apply Proposition 3.12 to the ρ^{-1}-dilate of $(\mathcal{T}_B, Y_B)$. However, the ρ^{-1}-dilate of $\mathcal{T}_B$ may not be a Katz–Tao $(\delta/\rho, t)$-set of tubes. To overcome this issue, we will randomly pick a $\mathcal{T}_B'' \subset \mathcal{T}_B$ that obeys that desired assumption. After that, we will apply Proposition 3.12 to $(\mathcal{T}_B'', Y_B)$. Let us turn to details.

Let $L_3(B) = \{\ell \in L_3 : Y_3(\ell) \cap B \neq \varnothing\}$, so $L_3(B) = \bigcup_{J \in \mathcal{T}_B} L_3(J)$. Let $\mu_J = \#L_3(J)$. By dyadic pigeonholing on $\{\mu_J : J \in \mathcal{T}_B\}$, there is a subset $\mathcal{T}_B' \subset \mathcal{T}_B$ and a uniform number μ_B such that the following is true:

(1) $\mu_J \sim \mu_B$ for all $J \in \mathcal{T}_B'$.
(2) $\mu_B \cdot \#\mathcal{T}_B' \gtrapprox \sum_{J \in \mathcal{T}_B} \mu_J \sim \#L_3(B)$.

For each $v \in [\delta, \rho]$, let $\mathcal{T}_v(B)$ be the family of distinct $v \times \rho$-tubes in B, each of which contains at least one $J \in \mathcal{T}'_B$. Define a multiplicity factor σ, which measures how much a set fails to be a Katz–Tao set, as follows:

$$\sigma := \sup_{v\in[\delta,\rho]} \{(\delta/v)^t \cdot \sup_{\tilde{J}\in\mathcal{T}_v(B)} \#\{J \subset \tilde{J} : J \in \mathcal{T}'_B\}\} \geqslant 1. \tag{3.37}$$

Note that for each $v \in [\delta, \rho]$ and each $\tilde{J} \in \mathcal{T}_v(B)$, the set of lines $L(\tilde{J}) = \{\ell \in L : |N_\delta(\ell) \cap J| \gtrsim |J| \text{ for some } J \subset \tilde{J},\ J \in \mathcal{T}'_B\}$ is contained in a $(v/\rho) \times 1$-tube. Since L is a Katz–Tao (δ, t)-set, $\#L(\tilde{J}) \lesssim (v/\delta\rho)^t$. Recall that there are $\gtrsim \#L_3(J) \sim \mu_B$ many $\ell \in L$ such that $|N_\delta(\ell) \cap J| \gtrsim |J|$ for each $J \in \mathcal{T}'_B$. Thus, for all $v \in [\delta, \rho]$ and all $\tilde{J} \in \mathcal{T}_v(B)$, $\mu_B^{-1}(v/\delta\rho)^t \gtrsim \mu_B^{-1}\#L(\tilde{J}) \gtrsim \#\{J \subset \tilde{J} : J \in \mathcal{T}'_B\}$. This implies

$$\sigma \cdot \mu_B \lesssim \rho^{-t}. \tag{3.38}$$

Let $\mathcal{T}''_B \subset \mathcal{T}'_B$ be a uniform random sample of probability σ^{-1}. Thus, with high probability, $\mathcal{T}''_B \gtrapprox \sigma^{-1}\mathcal{T}'_B$, and $\tilde{J}$ contains $\lessapprox (v/\delta)^t$ many $\delta \times \rho$-tubes in $\mathcal{T}''_B$ for all $\tilde{J} \in \mathcal{T}_v(B)$ and all $v \in [\delta, \rho]$. This shows that the ρ^{-1}-dilate of $\mathcal{T}''_B$ is a Katz–Tao $(\delta/\rho, t)$-set.

Recall that Y_B is a uniform, $\gtrapprox \lambda\rho^{-1+\varepsilon^2}$-dense shading, and the ρ^{-1}-dilate of $Y_B(J)$ is a $(\delta/\rho, \varepsilon^2, K')$-set for all $J \in \mathcal{T}_B$ and some $K' \lessapprox 1$. Apply the ρ-dilate version of Proposition 3.12 to $(\mathcal{T}''_B, Y_B)$ (with $\varepsilon/2$ in place of ε) so that

$$|E_{\mathcal{T}''_B, Y_B}| \gtrapprox c_\varepsilon(\delta/\rho)^{\varepsilon/2} \cdot (\lambda\rho^{-1+\varepsilon^2})^{1/2}(\delta/\rho)^{(t-1)/2}\gamma_{Y_B,t*}^{-1/2} \sum_{J\in\mathcal{T}''_B} |Y_B(J)|. \tag{3.39}$$

Recall that $\mathcal{T}''_B \gtrapprox \sigma^{-1}\mathcal{T}'_B$, $\#L_3(J) \sim \mu_B$ for all $J \in \mathcal{T}'_B$, and $\mu_B \cdot \#\mathcal{T}'_B \gtrapprox \#L_3(B)$. Since $|Y_B(J)|$ are the same up to a constant multiple, by (3.38),

$$\sum_{J\in\mathcal{T}''_B} |Y_B(J)| \gtrapprox \sigma^{-1} \sum_{J\in\mathcal{T}'_B} |Y_B(J)| \gtrapprox (\sigma\mu_B)^{-1} \sum_{\ell\in L_3(B)} |Y_3(\ell) \cap B| \tag{3.40}$$

$$\gtrapprox \rho^t \int_B \#L_3(x). \tag{3.41}$$

Since $\rho \geqslant \delta^{\varepsilon_1}$ and since $E_{\mathcal{T}''_B, Y_B} \subset E_L \cap B$, plug this back to (3.39) so that

$$|E_L \cap B| \gtrapprox c_\varepsilon(\delta/\rho)^{\varepsilon/2} \cdot (\lambda\rho^{-1+\varepsilon^2})^{1/2}(\delta/\rho)^{(t-1)/2}\gamma_{Y_B,t*}^{-1/2} \cdot \rho^t \int_B \#L_3(x) \tag{3.42}$$

$$\geqslant c_\varepsilon(\delta/\rho)^{\varepsilon/2} \cdot \rho^{\varepsilon^2/2}\delta^{t\varepsilon_1/2}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y_B,t*}^{-1/2} \int_B \#L_3(x). \tag{3.43}$$

Notice that $\gamma_{Y_B,t*} \lesssim \gamma_{Y,t*}$ for all $B \in \mathcal{B}_\rho$. Since $\mathcal{B}_\rho$ covers E_{L_3} and since $(L_3, Y_3)_\delta$ is a refinement of $(L, Y)_\delta$, we sum up all $B \in \mathcal{B}_\rho$ in (3.42) to get

$$|E_L| \gtrapprox c_\varepsilon\delta^{\varepsilon/2} \cdot \delta^{t\varepsilon_1/2}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t*}^{-1/2} \int \#L_3(x) \tag{3.44}$$

$$\geqslant c_\varepsilon\delta^{\varepsilon}\delta^{t\varepsilon_1/2}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t*}^{-1/2} \sum_{\ell\in L} |Y(\ell)|. \qquad \square$$

3.3. **Proof of the main proposition.** Finally, let us see how Proposition 3.10 and Lemma 3.7 imply Proposition 3.12.

Proof of Proposition 3.12. We will prove (3.34) by a backward induction on δ. In the base case when $\delta \gtrsim_\varepsilon 1$, we choose c_ε sufficiently small such that (3.34) is true. By dyadic pigeonholing, there exists a refinement L_1 of L such that for all $\ell \in L_1$, $|Y(\ell)|$ are the same up to a constant multiple. Without loss of generality, assume

$$|Y(\ell)| \sim \lambda\delta \tag{3.45}$$

for all $\ell \in L_1$. Denote by $Y_1 = Y$, so that $(L_1, Y_1)_\delta$ is a refinement of $(L, Y)_\delta$ and a family of δ-separated lines with a uniform, λ-dense shading. Moreover, there exists a $\rho \in [\delta, \delta^\eta]$ such that $Y_1(\ell)$ is a $(\delta, \varepsilon^2, C; \rho)$-set for all $\ell \in L_1$. Note that when $C \geqslant \delta^{-2\eta^2}$, $|E_L| \geqslant \sup_{\ell\in L}|Y(\ell)| \geqslant \delta^2 \geqslant C^{-\eta^{-2}}$, which implies (3.34) directly. Thus, we can assume $C \leqslant \delta^{-2\eta^2}$, in which case the spacing condition on Y_1 is not vacuous, as $\rho \leqslant \delta^\eta$.

Applying Lemma 2.11 and Lemma 2.13 to the set of shadings $\{Y_1(\ell)\}_{\ell\in L_1}$, we know that there is a refinement $(L_2, Y_2)_\delta$ of $(L_1, Y_1)_\delta$ such that $\{Y_2(\ell) : \ell \in L_2\}$ admits a uniform branching function. Let

$$\eta_1 := \eta\varepsilon^3. \tag{3.46}$$

Let $t^* = \min\{t, 2-t\}$. Apply Lemma 2.23 to this family of shadings with $\eta_0 = \eta_0(\eta_1)$ with $t = t^*$ to obtain a scale r such that the following is true:

(1) $r > \delta^{1-\eta_0\eta_1^{-1}}$.
(2) $\gamma_{\tilde{Y}_2,t^*}(\ell) \lesssim \gamma_{Y_2,t^*}(\ell)$, where $\tilde{Y}_2(\ell) = (Y_2(\ell))_r$.
(3) For each $\delta \times r$-tube $J \subset (Y_2(\ell))_r$, the r^{-1}-dilate of $Y_2(\ell) \cap J$ along ℓ is a $(\delta/r, s, (\delta/r)^{-9\eta_1})$-set, and $\log_{1/\delta}\left(\frac{|Y_2(\ell)|_\delta}{|Y_2(\ell)|_r}\right) \leqslant (s + 9\eta_1)\log_{1/\delta}(r/\delta)$.

Since $Y_2(\ell)$ is a refinement of $Y(\ell)$, by Lemma 2.9, $Y_2(\ell)$ is a $(\delta, \varepsilon^2, CC_2; \rho)$-set for some $C_2 \lessapprox 1$.

We consider two separate cases: $r \geqslant \delta^\eta$ and $r \leqslant \delta^\eta$. When $r \geqslant \delta^\eta$, note that for all $\ell \in L_2$, $Y_2(\ell)$ is a $(\delta, s, \delta^{-2\eta})$-set, $|Y_2(\ell)| \approx \lambda|N_\delta(\ell)|$, and $|Y_2(\ell)| \lesssim \delta^{1-s-2\eta}|N_\delta(\ell)|$. Recall that (ε, η) is chosen such that Lemma 3.7 is true for $(\nu, \eta) = (\varepsilon^2, 2\eta)$. Apply Lemma 3.7 to $(L_2, Y_2)_\delta$ to have

$$|E_{L_2}| \geqslant \delta^{\varepsilon^2+O(\eta)}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y_2,t^*}^{-1/2}\sum_{\ell\in L_2}|Y_2(\ell)|. \tag{3.47}$$

This concludes (3.34) as $(L_2, Y_2)_\delta$ is a refinement of $(L, Y)_\delta$.

From now on, we assume $r \leqslant \delta^\eta$ (we are only going to use this assumption in Step 5, Case **C**).

Step 1: Uniformization of E_L inside each r-ball

We will establish a uniformization on $E_L \cap B$ for each r-ball $B \subset (E_L)_r$. Define

$$\tilde{\lambda} := \lambda(r/\delta)^{1-s-\eta_1}. \tag{3.48}$$

Let $\mathcal{T}_{\delta/r}$ be a maximal collection of distinct $(\delta/r) \times 1$-tubes. For each $T \in \mathcal{T}_{\delta/r}$, recall Definition 3.2 for $L_2[T]$, and let $\tilde{L}_2[T] = L_2[T]$. For each $\ell \in L_2$, define a new shading $\tilde{Y}_2(\ell) = (Y_2(\ell))_r$ by $\delta\times r$-tubes as in Definition 2.18. Note that for each $\delta \times r$-tube $J \subset \tilde{Y}_2(\ell)$, we have $|Y_2(\ell) \cap J| \lesssim (\delta/r)^{1-s-\eta_1}|N_\delta(\ell) \cap J|$. Consequently,

$\tilde{Y}_2(\ell)$ is $\gtrsim \tilde{\lambda}$-dense, as $Y_2(\ell)$ is $\gtrsim \lambda$-dense. Apply Lemma 2.24 to $(\tilde{L}_2[T], \tilde{Y}_2)_\delta$ to obtain a refinement $(\tilde{L}_2'[T], \tilde{Y}_2')_\delta$ of $(\tilde{L}_2[T], \tilde{Y}_2)_\delta$ satisfying

$$\#(\tilde{L}_2[T])_{\tilde{Y}_2'}(x) \sim \frac{\sum_{\ell\in\tilde{L}_2[T]} |\tilde{Y}_2(\ell)|}{|E_{\tilde{L}_2[T],\tilde{Y}_2'}|} \approx \frac{\sum_{\ell\in\tilde{L}_2'[T]} |\tilde{Y}_2'(\ell)|}{|E_{\tilde{L}_2'[T]}|}. \tag{3.49}$$

Note that $\tilde{Y}_2'(\ell)$ is $\gtrsim \tilde{\lambda}$-dense and is still a union of $\delta \times r$-tubes.

Recall that $Y_2(\ell)$ is a $(\delta, \varepsilon^2, CC_2; \rho)$-set for some $C_2 \lesssim 1$. Since Y_2 is uniform, $\tilde{Y}_2(\ell)$ is a $(r, \varepsilon^2, C\tilde{C}_2; \max\{\rho, r\})$-set for some $\tilde{C}_2 \lesssim 1$. Thus, the anisotropic r/δ-dilate of $(\tilde{L}_2'[T], \tilde{Y}_2')_\delta$ is a $(r, \varepsilon^2, C\tilde{C}_2; \max\{\rho, r\})$-set. Clearly, $\max\{\rho, r\} \in [r, r^\eta]$, as $\rho \in [\delta, \delta^\eta]$. Apply Proposition 3.12 at scale r to this anisotropic r/δ-dilate to bound $E_{\tilde{L}_2'[T]}$ so that

$$\mu_T := \frac{\sum_{\ell\in\tilde{L}_2'[T]} |\tilde{Y}_2'(\ell)|}{|E_{\tilde{L}_2'[T]}|} \lesssim c_\varepsilon^{-1} r^{-\varepsilon} (C\tilde{C}_2)^{\eta^{-2}} \tilde{\lambda}^{-1/2} r^{(1-t)/2} \gamma_{\tilde{Y}_2', t*}^{1/2} \tag{3.50}$$

$$\leqslant c_\varepsilon^{-1} r^{-\varepsilon} (C\tilde{C}_2)^{\eta^{-2}} \tilde{\lambda}^{-1/2} r^{(1-t)/2} \gamma_{Y_2, t*}^{1/2}. \tag{3.51}$$

We use the estimate $\gamma_{\tilde{Y}_2', t*} \leqslant \gamma_{\tilde{Y}_2, t*} \lesssim \gamma_{Y_2, t*}$ in the last inequality. Since the intersection of $N_\delta(\ell_1)$ and $N_\delta(\ell_2)$ contains at least one $\delta \times r$-tube for any two $\ell_1, \ell_2 \in L_2[T]$ with $\tilde{Y}_2'(\ell_1) \cap \tilde{Y}_2'(\ell_2) \neq \varnothing$, the above estimate shows that for each $\delta \times r$-tube $J \subset E_{\tilde{L}_2(T), \tilde{Y}_2'}$ parallel to T,

$$\#\{\ell \in \tilde{L}_2[T], J \subset \tilde{Y}_2'(\ell)\} \approx \mu_T \lesssim c_\varepsilon^{-1} r^{-\varepsilon} (C\tilde{C}_2)^{\eta^{-2}} \tilde{\lambda}^{-1/2} r^{(1-t)/2} \gamma_{Y_2, t*}^{1/2} \tag{3.52}$$

Let $\tilde{L}_2 = \bigcup_{T\in\mathcal{T}_{\delta/r}} \tilde{L}_2[T]$ and let $Y_3'(\ell) = \tilde{Y}_2'(\ell) \cap Y_2(\ell)$ for any $\ell \in \tilde{L}_2'$. Since $|Y_2(\ell) \cap J|$ are about the same up to a constant multiple, $(\tilde{L}_2, Y_3')_\delta$ is a refinement of $(L_2, Y_2)_\delta$. By dyadic pigeonholing, there exists a uniform

$$\mu_3 \lesssim c_\varepsilon^{-1} r^{-\varepsilon} (C\tilde{C}_2)^{\eta^{-2}} \tilde{\lambda}^{-1/2} r^{(1-t)/2} \gamma_{Y_2, t*}^{1/2} \tag{3.53}$$

and a refinement $(L_3, Y_3')_\delta$ of $(\tilde{L}_2, Y_3')_\delta$ such that the following holds. For any $T \in \mathcal{T}_{\delta/r}$ with $L_3[T]$ nonempty, we have $\mu_T \sim \mu_3$. It follows from (3.52) that for any $\delta \times r$-tube $J \subset T$ parallel to T and $E_{L_3[T], Y_3'} \cap J \neq \varnothing$,

$$\#\{\ell \in L_3[T] : Y_3'(\ell) \cap J \neq \varnothing\} \approx \mu_3. \tag{3.54}$$

Note that for each r-ball B, $\tilde{Y}_2'(\ell) \cap B$ is either an empty set, or is essentially an $\delta \times r$-tube. Let $\mathcal{B}_{r,3}$ be the set of finite-overlapping r-balls contained in $(E_{L_3, Y_3'})_r$. For each $B \in \mathcal{B}_{r,3}$, consider the family of distinct $\delta \times r$-tubes

$$\mathcal{T}_B^0 := \{J : \tilde{Y}_2'(\ell) \cap B \subset J \text{ for some } \ell \in L_3\}. \tag{3.55}$$

Thus, if $Y_3'(\ell) \cap B \neq \varnothing$, then there exists a $\delta \times r$-tube $J \in \mathcal{T}_B^0$ such that $Y_3'(\ell) \cap B \subset J$. By Lemma 2.11, there is a uniform refinement $\mathcal{T}_B$ of $\mathcal{T}_B^0$.

Now for each $\ell \in L_3$, define a new shading $Y_3(\ell) \subset Y_3'(\ell)$ as follows: First, for each r-ball $B \in \mathcal{B}_{r,3}$ such that $Y_3'(\ell) \cap B \neq \varnothing$, define

$$Y_3(\ell) \cap B = \begin{cases} Y_3'(\ell) \cap B, & \text{if } \exists J \in \mathcal{T}_B \text{ such that } \tilde{Y}_2'(\ell) \cap B \subset J; \\ \varnothing, & \text{otherwise.} \end{cases} \tag{3.56}$$

Then, define $Y_3(\ell) = \bigcup_{B\in\mathcal{B}_{r,3}} Y_3(\ell) \cap B$.

Since $\mathcal{T}_B$ is a refinement of $\mathcal{T}_B^0$, it follows from (3.54) that for each $B \in \mathcal{B}_{r,3}$,

$$\int_B \#(L_3)_{Y_3'}(x) \lessapprox \int_B \#(L_3)_{Y_3}(x). \tag{3.57}$$

This shows that $(L_3, Y_3)_\delta$ is a refinement of $(L_3, Y_3')_\delta$.

Since $Y_3(\ell)$ is a refinement of $Y_2(\ell)$ at resolution r: for all $\ell \in L_3$ and all $\delta \times r$-tubes $J \subset (Y_3(\ell))_r$, $Y_3(\ell) \cap J = Y_2(\ell) \cap J$, the r^{-1}-dilate of $Y_3(\ell) \cap J$ along ℓ is a $(\delta/r, s, (\delta/r)^{-\eta_1})$-set, and $|Y_3(\ell) \cap J|$ are about the same up to a constant multiple.

As a remark, we remind the reader that the following is true: For all $\ell \in L_3$ and all $\delta \times r$-tubes $J \subset (Y_3(\ell))_r$, we have

(1) $Y_3(\ell) \cap J = Y_2(\ell) \cap J$.
(2) The r^{-1}-dilate of $Y_3(\ell) \cap J$ along ℓ is an $(\delta/r, s, (\delta/r)^{-\eta_1})$-set.
(3) $|Y_3(\ell) \cap J| \lesssim (\delta/r)^{1-s-\eta_1}|J|$.
(4) $|Y_3(\ell) \cap J|$ are about the same.

Moreover, for each $B \in \mathcal{B}_{r,3}$, there is a uniform set of $\delta \times r$-tubes $\mathcal{T}_B$ such that for all $\ell \in L_3$, $Y_3(\ell) \cap B \subset J$ for some $J \in \mathcal{T}_B$. Also, by (3.54), for all $J \in \mathcal{T}_B$, we have

$$\#\{\ell \in L_3 : J \subset (Y_3(\ell))_r\} \approx \mu_3. \tag{3.58}$$

Step 2: A broad-narrow argument.

Applying Lemma 2.25 to $(L_3, Y_3)_\delta$ (with α in place of ρ), after dyadic pigeonholing, there is an $\alpha \in [\delta, 1]$ and a set $E_\alpha \subset E_{L_3}$ such that

(1) For each $x \in E_\alpha$, there exists a 10α-cap $\sigma_x \subset \mathbb{S}^1$ and a refinement $L_3'(x)$ of $L_3(x)$ such that the direction $V(\ell) \in \sigma_x$ for all $\ell \in L_3'(x)$.
(2) There are two subsets of lines $L', L'' \subset L_3'(x)$ such that $\#L', \#L'' \gtrapprox \#L_3'(x)$, and $\alpha \geqslant \measuredangle(\ell', \ell'') \gtrapprox \alpha$ for all $\ell' \in L', \ell'' \in L''$.

Define for each $\ell \in L_3$ a new shading $Y_4(\ell) = E_\alpha \cap Y_3(\ell)$, so $(L_3, Y_4)_\delta$ is a refinement of $(L_3, Y_3)_\delta$. Since $(L_3, Y_3)_\delta$ is a refinement of $(L_2, Y_2)_\delta$, $(L_3, Y_4)_\delta$ is a refinement of $(L_2, Y_2)_\delta$. Thus, by dyadic pigeonholing, there exists a refinement $(L_4, Y_4)_\delta$ of $(L_3, Y_4)_\delta$ (so it is a refinement of $(L_2, Y_2)_\delta$ as well) such that $Y_4(\ell)$ is a refinement of $Y_2(\ell)$ for all $\ell \in L_4$.

Suppose $\alpha \leqslant (\delta/r)^{\eta_1}$. Recall that $Y_2(\ell)$ is uniform for all $\ell \in L_2 \supset L_4$. Since for each $\ell \in L_4$, $Y_4(\ell)$ is a refinement of $Y_2(\ell)$, by Lemma 2.11, there exists a new shading Y_4' such that $Y_4'(\ell)$ is uniform and is a refinement of $Y_4(\ell)$ for all $\ell \in L_4$. For each $\ell \in L_4$, consider a new shading $\tilde{Y}_4'(\ell) = (Y_4'(\ell))_{\delta/\alpha}$. Since $Y_4'(\ell)$ is uniform, there exists a $\kappa(\ell) < 1$ such that the following is true:

(1) $|Y_4'(\ell)| \approx \kappa(\ell)|\tilde{Y}_4'(\ell)|$.
(2) $|Y_4'(\ell) \cap J| \approx \kappa(\ell)|J|$ for each $\delta \times (\delta/\alpha)$-tube $J \subset (Y_4'(\ell))_{\delta/\alpha}$.

By dyadic pigeonholing on $\{\kappa(\ell) : \ell \in L_4\}$, there exists a uniform $\kappa < 1$ and a set of lines $L_4' \subset L_4$ such that the following is true.

(1) $(L_4', Y_4')_\delta$ is a refinement of $(L_4, Y_4)_\delta$.
(2) $\kappa(\ell) \sim \kappa \leqslant 1$ for all $\ell \in L_4'$.

Let $\mathcal{T}_\alpha$ be a collection of α-separated $\alpha \times 1$-tubes, and for each $T \in \mathcal{T}_\alpha$, recall Definition 3.2 for $L_4'[T]$. Then

$$|E_{L_4}| \gtrsim \sum_{T \in \mathcal{T}_\alpha} |E_{L_4'[T]}| \tag{3.59}$$

For each $T \in \mathcal{T}_\alpha$, let ϕ_T be the anisotropic α^{-1}-dilate which maps T to the unit ball, so $|\phi_T(E_{L_4'[T]})| = (\alpha/\delta)|E_{L_4[T]}|$. For all $\ell \in L_4' \subset L_2$, recall that $Y_2(\ell)$ is a $(\delta, \varepsilon^2, CC_2; \rho)$-set for some $C_2 \lessapprox 1$. Since $Y_4'(\ell)$ is a refinement of $Y_2(\ell)$, $Y_4'(\ell)$ is a $(\delta, \varepsilon^2, CC_4; \rho)$-set for some $C_4 \lessapprox 1$. Thus, $\tilde{Y}_4'(\ell)$ is a $(\delta/\alpha, \varepsilon^2, CC_4; \max\{\rho, \delta/\alpha\})$-set, implying that $\phi_T(\tilde{Y}_4')$ is a $(\delta/\alpha, \varepsilon^2, CC_4; \max\{\rho, \delta/\alpha\})$-set. Clearly, $\max\{\rho, \delta/\alpha\} \in [\delta/\alpha, (\delta/\alpha)^\eta]$, as $\rho \in [\delta, \delta^\eta]$. Moreover, since $Y_4'(\ell)$ is $\gtrapprox \lambda$-dense, we have that $\tilde{Y}_4'(\ell)$ is $\gtrapprox \lambda\kappa^{-1}$-dense, which implies $\phi_T(\tilde{Y}_4')$ is $\gtrapprox \lambda\kappa^{-1}$-dense. Apply induction at scale δ/α to $(\phi_T(L_4'[T]), \phi_T(\tilde{Y}_4'))_{\delta/\alpha}$ so that, after rescaling back, we have

$$|E_{L_4'[T], \tilde{Y}_4'}| \geqslant c_\varepsilon (\delta/\alpha)^\varepsilon (CC_4)^{-\eta^{-2}} \lambda^{1/2} \kappa^{-1/2} (\delta/\alpha)^{(t-1)/2} \gamma_{\tilde{Y}_4', t^*}^{-1/2} \sum_{\ell \in L_4'[T]} |\tilde{Y}_4'(\ell)| \tag{3.60}$$

$$\gtrapprox c_\varepsilon (\delta/\alpha)^\varepsilon (CC_4)^{-\eta^{-2}} \lambda^{1/2} \kappa^{-1} \delta^{(t-1)/2} \gamma_{Y_4', t^*}^{-1/2} \sum_{\ell \in L_4'[T]} |Y_4'(\ell)|. \tag{3.61}$$

In the last inequality, we use $\gamma_{\tilde{Y}_4', t^*} \leqslant \alpha^{1-t^*} \kappa^{-1} \gamma_{Y_4', t^*}$, a consequence of Lemma 2.22 (note that when $t > 1$, $\alpha^{1-t^*} = \alpha^{t-1}$).

Since $|E_{L_4[T]}| \gtrsim \kappa |E_{L_4'[T], \tilde{Y}_4'}|$, we thus have

$$|E_{L_4[T]}| \gtrapprox c_\varepsilon (\delta/\alpha)^\varepsilon (CC_4)^{-\eta^{-2}} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y_4', t^*}^{-1/2} \sum_{\ell \in L_4'[T]} |Y_4'(\ell)|. \tag{3.62}$$

Finally, since $(L_4', Y_4')_\delta$ is a refinement of $(L, Y)_\delta$ and since $\gamma_{Y_4', t^*} \leqslant \gamma_{Y, t^*}$, by (3.59), the above estimate implies

$$|E_L| \geqslant |E_{L_4}| \geqslant c_\varepsilon \delta^\varepsilon C^{-\eta^{-2}} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y, t^*}^{-1/2} \sum_{\ell \in L} |Y(\ell)|. \tag{3.63}$$

Suppose $\alpha \geqslant (\delta/r)^{\eta_1}$. We proceed to the next step.

Step 3: Setting up an incidence problem inside an r-ball.

For $k = 3, 4$ and each $\ell \in L_k$, let $\mathcal{J}_k(\ell) = \{J \subset (Y_k(\ell))_r\}$. Recall that $Y_4(\ell)$ is a $\gtrapprox 1$-refinement of $Y_2(\ell)$. Since $Y_4(\ell) \subset Y_3(\ell) \subset Y_2(\ell)$ and since $Y_2(\ell)$ is uniform, we have $\#\mathcal{J}_4(\ell) \leqslant \#\mathcal{J}_3(\ell) \lessapprox \#\mathcal{J}_4(\ell)$.

Let $\mathcal{J}_5(\ell) = \{J \in \mathcal{J}_4(\ell) : |Y_4(\ell) \cap J| \gtrapprox |Y_3(\ell) \cap J|\}$. Recall that $|Y_3(\ell) \cap J| = |Y_2(\ell) \cap J|$ are the same up to a constant multiple for all $J \in \mathcal{J}_3(\ell)$. Thus, since $Y_4(\ell)$ is a $\gtrapprox 1$-refinement of $Y_3(\ell)$, we have $\#\mathcal{J}_5(\ell) \gtrapprox \#\mathcal{J}_4(\ell)$. Consider a new shading $Y_5(\ell) = \bigcup_{J \in \mathcal{J}_5(\ell)} J \cap Y_4(\ell)$, so $Y_5(\ell)$ is a $\gtrapprox 1$-refinement of $Y_4(\ell)$. Consequently, by taking $L_5 = L_4$, $(L_5, Y_5)_\delta$ is a refinement of $(L_4, Y_4)_\delta$.

We remark that the set of lines and shading $(L_5, Y_5)_\delta$ has the following properties

1. For any $x \in E_{L_5}$, $(L_3)_{Y_3}(x)$ is two-broad (i.e. $\alpha \geqslant (\delta/r)^{\eta_1}$, assumed in Step 2), since $E_{L_5} \subset E_{L_4} \subset E_\alpha$.
2. For any $\ell \in L_5$ and any $\delta \times r$-tube $J \in \mathcal{J}_5(\ell)$, the (1-dimensional) r^{-1}-dilate of $Y_5(\ell) \cap J$ along the line ℓ is a $(\delta/r, s, (\delta/r)^{-2\eta_1})$-set. This is a consequence of Lemma 2.6 and the fact that the r^{-1}-dilate of $Y_3(\ell) \cap J$ is a $(\delta/r, s, (\delta/r)^{-\eta_1})$-set (see the end of Step 1).

Let $\mathcal{B}_{r,5} \subset \mathcal{B}_{r,3}$ be the family of r-balls contained in $(E_{L_5, Y_5})_r$. For each r-ball $B \in \mathcal{B}_{r,5} \subset \mathcal{B}_{r,3}$, let $\mathcal{T}_B$ be the set of distinct $\delta \times r$-tube segments defined in the end of Step 1. We restate some properties of $\mathcal{T}_B$ here:

1. For each $\ell \in L_3$ with $Y_3(\ell) \cap B \neq \varnothing$, there exists $J \in \mathcal{T}_B$ such that $J \subset (Y_3(\ell))_r$.

(2) For any two $J, J' \in \mathcal{T}_B$, $|J \cap J'| \leqslant |J|/2$.
(3) $\mathcal{T}_B$ is uniform, and for each $J \in \mathcal{T}_B$, (3.58) is true.

We are going to analyze the incidences between $\delta \times r$-tubes in $\mathcal{T}_B$ and δ-balls in $B \cap E_{L_3,Y_3}$ (here we choose E_{L_3,Y_3} over E_{L_5,Y_5} because of the following: Each $x \in E_{L_5,Y_5}$ is "two-broad" with respect to $(L_3, Y_3)_\delta$ but not with respect to $(L_5, Y_5)_\delta$).

First, we study the structure of $\mathcal{T}_B$. Pick $\ell \in L_5$ such that $Y_5(\ell) \cap B \neq \varnothing$, so the r^{-1}-dilate of $Y_5(\ell) \cap B$ along the line ℓ is a $(\delta/r, s, (\delta/r)^{-2\eta_1})$-set. Recall the "two-broad" property obtained in Step 2: For each $x \in E_{L_5} \subset E_\alpha$, there exists $\ell' \in L_3$ such that $x \in Y_3(\ell')$ and $\angle(\ell, \ell') \gtrapprox \alpha \geqslant (\delta/r)^{\eta_1}$. Hence, for each $v \in [\delta, r]$, the number of distinct $v \times r$-tubes required to cover $\mathcal{T}_B$ is $\gtrsim (r\alpha/v)^{s-2\eta_1} \geqslant (r/v)^s(\delta/r)^{3\eta_1}$. Let $\mathcal{T}_v(B)$ denote this set of $v \times r$-tubes.

Next, for each $J \in \mathcal{T}_B$, define

$$L_3(J) := \{\ell \in L_3 : J \subset (Y_3(\ell))_r\}, \tag{3.64}$$

and let $L_3(B) = \bigsqcup_{J\in\mathcal{T}_B} L_3(J)$ be the set of lines whose shading intersects B. Define $K_J = \#L_3(J)$, so by (3.58), $K_J \approx \mu_3$. Let $\ell_1(J), \ldots, \ell_{K_J}(J)$ be an enumeration of the lines in $L_3(J)$. Such enumeration on each $L_3(J)$ gives a natural disjoint partition of $L_3(B)$: Let $L_{3,k}(B) = \{\ell_k(J) : J \in \mathcal{T}_B\}$, where $\ell_k(J) := \varnothing$ if $k > K_J$. For each k and each $J \in \mathcal{T}_B$, define a shading

$$Y_{B,k}(J) = \begin{cases} Y_3(\ell_k(J)) \cap J, & \text{if } k \leqslant K_J \\ \varnothing, & \text{otherwise.} \end{cases} \tag{3.65}$$

Let $K_B = \min_{J\in\mathcal{T}_B} K_J \approx \mu_3$, so when $k \leqslant K_B$, $Y_{B,k}(J) \neq \varnothing$ for all $J \in \mathcal{T}_B$. Since $|Y_3(\ell) \cap J|$ are about the same for all $\ell \in L_3(J)$ and all $J \subset (Y_3(\ell))_r$, we know that $\sum_{J\in\mathcal{T}_B} |Y_{B,k}(J)| \approx \sum_{J\in\mathcal{T}_B} |Y_{B,k'}(J)|$ when $k, k' \leqslant K_B$, and $\sum_{J\in\mathcal{T}_B} |Y_{B,k}(J)| \gtrapprox \sum_{J\in\mathcal{T}_B} |Y_{B,k'}(J)|$ when $k \leqslant K_B \leqslant k'$. Notice that $Y_{B,k} = \varnothing$ whenever $k > \max_{J\in\mathcal{T}_B} K_J \approx \mu_3 \approx K_B$. Consequently,

$$\int_B \#L_3(x) = \sum_k \sum_{p\subset E_{\mathcal{T}_B,Y_{B,k}}} \#\mathcal{T}_B(p) \lessapprox \sum_{k\leqslant K_B} \sum_{p\subset E_{\mathcal{T}_B,Y_{B,k}}} \#\mathcal{T}_B(p). \tag{3.66}$$

Step 4.1: Analyze $E_L \cap B$ for each r-ball B in the broad case (I).

Recall that for all $B \in \mathcal{B}_{r,5}$ and all $k \leqslant K_B$, the configuration $(\mathcal{T}_B, Y_{B,k})$ satisfies the following properties:

(1) $\mathcal{T}_B$ is uniform.
(2) For each $v \in [\delta, r]$, there are $\gtrsim (r/v)^s(\delta/r)^{3\eta_1}$ distinct $v \times r$-tubes containing at least one $J \in \mathcal{T}_B$.
(3) For each $J \in \mathcal{T}_B$, $|Y_{B,k}(J)| \lesssim (\delta/r)^{1-s-\eta}|J|$, and the one-dimensional r^{-1}-dilate of $Y_{B,k}(J)$ along J is a $(\delta/r, s, (\delta/r)^{-\eta_1})$-set.

We want to apply Proposition 3.10 to the r^{-1}-dilate of $(\mathcal{T}_B, Y_{B,k})$. However, the r^{-1}-dilate of $(\mathcal{T}_B, Y_{B,k})$ may not obey its Assumption (iii). To get around this issue, we apply Lemma 2.24 to $(\mathcal{T}_B, Y_{B,k})$ to obtain a refinement $(\mathcal{T}'_{B,k}, Y'_{B,k})$ of $(\mathcal{T}_B, Y_{B,k})$ such that

(1) For any $J \in \mathcal{T}'_{B,k}$, the one-dimensional r^{-1}-dilate of $Y'_{B,k}(J)$ along J is a $(\delta/r, s, (\delta/r)^{-2\eta_1})$-set (achieved by Lemma 2.6).

(2) Let $E_{\mathcal{T}'_{B,k},Y'_{B,k}} = \bigcup_{T\in\mathcal{T}'_{B,k}} Y'_{B,k}(J)$. For any δ-ball $p \subset E_{\mathcal{T}'_{B,k},Y'_{B,k}}$, $\mathcal{T}'_{B,k}(p) := \{J \in \mathcal{T}'_{B,k} : p \subset Y'_{B,k}(J)\}$ satisfies

$$\#\mathcal{T}'_{B,k}(p) \lessapprox |E_{\mathcal{T}'_{B,k},Y'_{B,k}}|^{-1} \sum_{J\in\mathcal{T}'_{B,k}} |Y'_{B,k}(J)|. \tag{3.67}$$

Since $|Y_{B,k}(J)|$ are about the same for all $J \in \mathcal{T}_B$, we have $\#\mathcal{T}'_{B,k} \gtrapprox \#\mathcal{T}_B$. Apply Lemma 2.11 to $\mathcal{T}'_{B,k}$ so that there is a uniform refinement $\mathcal{T}''_{B,k}$ of $\mathcal{T}'_{B,k}$. For each $J \in \mathcal{T}_B$, let $Y''_{B,k}(J) = Y'_{B,k}(J)$ if $J \in \mathcal{T}''_{B,k}$; otherwise, let $Y''_{B,k}(J) = \varnothing$.

Recall that the parameter η is chosen such that Proposition 3.10 is true when $(\nu,\eta) = (\varepsilon^2, 2\eta)$. Since $(\mathcal{T}''_{B,k}, Y''_{B,k})$ is a refinement of $(\mathcal{T}_B, Y_{B,k})$ and since the uniform set $\mathcal{T}''_{B,k}$ is also a refinement of the uniform set $\mathcal{T}_B$, we know that

(1) For each $J \in \mathcal{T}''_{B,k}$, the one-dimensional r^{-1}-dilate of $Y''_{B,k}(J)$ along the tube segment J is a $(\delta/r, s, (\delta/r)^{-2\eta_1})$-set.
(2) For each $v \in [\delta, r]$, there are $\gtrsim (r/v)^s(\delta/r)^{3\eta_1}$ distinct $v\times r$-tubes in $\mathcal{T}_v(B)$ containing at least a tube $J \in \mathcal{T}''_{B,k}$.
(3) We have $\#\mathcal{T}''_{B,k}(p) \leqslant \#\mathcal{T}'_{B,k}(p) \lessapprox |E_{\mathcal{T}'_{B,k},Y'_{B,k}}|^{-1} \sum_{J\in\mathcal{T}'_{B,k}} |Y'_{B,k}(J)| \lessapprox |E_{\mathcal{T}''_{B,k},Y''_{B,k}}|^{-1} \sum_{J\in\mathcal{T}''_{B,k}} |Y''_{B,k}(J)|$ for all $p \subset E_{\mathcal{T}''_{B,k},Y''_{B,k}}$.

Now we apply Proposition 3.10 to the r^{-1}-dilate of $(\mathcal{T}''_{B,k}, Y''_{B,k})$. After rescaling back, there is a refinement $(\mathcal{T}'''_{B,k}, Y'''_{B,k})$ of $(\mathcal{T}''_{B,k}, Y''_{B,k})$ and three possible outcomes:

A. Define $\mu_{B,k} := \max_{p\subset E_{\mathcal{T}'''_{B,k},Y'''_{B,k}}} \#\mathcal{T}'''_{B,k}(p)$, then $\mu_{B,k} \lessapprox (\delta/r)^{(s-t)/2}$.

B. We have

$$|E_{\mathcal{T}'''_{B,k},Y'''_{B,k}}| \gtrapprox (\delta/r)^{\varepsilon^2+3\eta}(\delta/r)^{2-(t+3s)/2}|B|. \tag{3.68}$$

C. There exists a scale $\Delta_B \gtrsim r(\delta/r)^{1-\sqrt{\eta}}$, and for each $J \in \mathcal{T}'''_{B,k}$, two numbers $d_{B,1}(J), d_{B,2}(J)$ and two families of disjoint Δ_B-balls $\mathcal{Q}_{B,1}(J), \mathcal{Q}_{B,2}(J)$ such that the following is true:
   (a) $\#\mathcal{Q}_{B,j}(J) = d_{B,j}(J), j = 1, 2$, and $\cup_{\mathcal{Q}_{B,2}(J)} \subset \cup_{\mathcal{Q}_{B,1}(J)} \subset N_{\Delta_B}(J)$.
   (b) $|E_{\mathcal{T}''_{B,k},Y''_{B,k}} \cap Q| \gtrsim (\delta/r)^{1-s+2\eta} d_{B,1}(J)^{-1}(r/\Delta_B)|Q|$ for all $Q \in \mathcal{Q}_{B,1}(J)$.
   (c) For each $Q \in \mathcal{Q}_{B,2}(J)$, Q contains $\gtrsim (\delta/r)^{-s-\eta} d_{B,2}(J)^{-1}$ many δ-balls in $Y'''_{B,k}(J)$.

For each $k \leqslant K_B$, one of the above outcome happens for $L_{3,k}(B)$. For $\mathbf{X} \in \{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$, let $\mathcal{K}_B(\mathbf{X})$ be the set of k such that outcome $\mathbf{X}$ holds for $L_{3,k}(B)$. Since $(\mathcal{T}'''_{B,k}, Y'''_{B,k})$ is a refinement of $(\mathcal{T}_B, Y_{B,k})$ for each k, we know from (3.66) that

$$\int_B \#L_3(x) = \sum_{k\leqslant K_B} \sum_{p\subset E_{\mathcal{T}_B,Y_{B,k}}} \#\mathcal{T}_B(p) \lessapprox \sum_{k\leqslant K_B} \sum_{p\subset E_{\mathcal{T}'''_{B,k},Y'''_{B,k}}} \#\mathcal{T}'''_{B,k}(p). \tag{3.69}$$

By pigeonholing, there exists an $\mathbf{X} \in \{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$ such that

$$\int_B \#L_3(x) \lessapprox \sum_{k\in\mathcal{K}_B(\mathbf{X})} \sum_{p\subset E_{\mathcal{T}'''_{B,k},Y'''_{B,k}}} \#\mathcal{T}'''_{B,k}(p). \tag{3.70}$$

We remark that (3.70) applies to each $B \in \mathcal{B}_{r,5}$.

Step 4.2: Analyze $E_L \cap B$ for each r-ball B in the broad case (II).

Since $(L_5, Y_5)_\delta$ is a refinement of $(L_3, Y_3)_\delta$ and since for each $B \in \mathcal{B}_{r,5}$, $B \cap Y_5(\ell) \neq \varnothing$ for some $\ell \in L_5$, we know that

$$\int_{E_{L_3}} \#L_3(x) \lessapprox \int_{E_{L_5}} \#L_5(x) = \sum_{B \subset \in \mathcal{B}_{r,5}} \int_B \#L_5(x) \leqslant \sum_{B \in \mathcal{B}_{r,5}} \int_B \#L_3(x). \tag{3.71}$$

By (3.70) and pigeonholing, there exists a $\mathbf{X} \in \{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$ and a subset $\mathcal{B}_{r,5}(\mathbf{X}) \subset \mathcal{B}_{r,5}$ such that

$$\int_{E_{L_3}} \#L_3(x) \lessapprox \sum_{B \in \mathcal{B}_{r,5}(\mathbf{X})} \sum_{k \in \mathcal{K}_B(\mathbf{X})} \sum_{p \subset E_{\mathcal{T}'''_{B,k}, Y'''_{B,k}}} \#\mathcal{T}'''_{B,k}(p). \tag{3.72}$$

The configurations $\{(\mathcal{T}'''_{B,k}, Y'''_{B,k}) : B \in \mathcal{B}_{r,5}(\mathbf{X}), k \in K_B(\mathbf{X})\}$ define a new shading Y_6 for each $\ell \in L_3$ as follows: Let $\mathcal{B}_{r,5}(\ell) \subset \mathcal{B}_{r,5}(\mathbf{X})$ be the family of r-balls obeying $Y_3(\ell) \cap B \neq \varnothing$. For each $B \in \mathcal{B}_{r,5}(\ell)$, we know that

(1) There is a $k \in \mathcal{K}_B(\mathbf{X})$ such that $\ell \in L_{3,k}(B)$.
(2) For this k, there is a $J \in \mathcal{T}_B$ such that $Y_{B,k}(J) \supset Y_3(\ell) \cap B$.

Now, for this $J \in \mathcal{T}_B$, we define

$$Y_6(\ell) \cap B := Y'''_{B,k}(J), \quad \text{and} \quad Y_6(\ell) = \bigcup_{B \in \mathcal{B}_{r,5}(\ell)} Y_6(\ell) \cap B. \tag{3.73}$$

Let $L_6 = L_3$. Then by (3.72), $(L_6, Y_6)_\delta$ is a refinement of $(L_3, Y_3)_\delta$.

The definition of $(L_6, Y_6)_\delta$ is subject to the outcome $\mathbf{X} \in \{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$ from pigeonholing. Let us discuss what happens for different possibilities of $\mathbf{X}$. Define $\mathcal{B}_{r,6} = \mathcal{B}_{r,5}(\mathbf{X})$, so $E_{L_6} \subset \cup_{\mathcal{B}_{r,6}}$.

When $\mathbf{X} = \mathbf{A}$: For each $B \in \mathcal{B}_{r,6}$, we have $\mu_{B,k} \lessapprox (\delta/r)^{(s-t)/2}$ for each $k \in \mathcal{K}_B(\mathbf{X})$. As a result, for each $x \in E_{L_6} \cap B$, since $\#\mathcal{K}_B(\mathbf{X}) \lessapprox \mu_3$,

$$\#L_6(x) \leqslant \sum_{k \in \mathcal{K}_B(\mathbf{X})} \mu_{B,k} \lessapprox \#\mathcal{K}_B(\mathbf{X}) \cdot \mu_{B,k} \lessapprox \mu_3 \cdot (\delta/r)^{(s-t)/2}. \tag{3.74}$$

When $\mathbf{X} = \mathbf{B}$: For each $B \in \mathcal{B}_{r,6}$, since $E_{\mathcal{T}'''_{B,k}, Y'''_{B,k}} \subset E_{L_6} \cap B$ for any $k \in \mathcal{K}_B(\mathbf{X})$,

$$|E_{L_6} \cap B| \geqslant \max_{k \in \mathcal{K}_B(\mathbf{X})} |E_{\mathcal{T}'''_{B,k}, Y'''_{B,k}}| \gtrapprox (\delta/r)^{\varepsilon^2 + 3\eta} (\delta/r)^{2-(t+3s)/2} |B|. \tag{3.75}$$

When $\mathbf{X} = \mathbf{C}$: For each $B \in \mathcal{B}_{r,6}$ and each $k \in \mathcal{K}_B(\mathbf{X})$, there exists a scale $\Delta_B \gtrsim r(\delta/r)^{1-\sqrt{\eta}}$, and for each $J \in \mathcal{T}'''_{B,k}$, two numbers $d_{B,1}(J), d_{B,2}(J)$ and two families of disjoint Δ_B-balls $\mathcal{Q}_{B,1}(J), \mathcal{Q}_{B,2}(J)$ such that the following is true:

(1) $\#\mathcal{Q}_{B,j}(J) = d_{B,j}(J), j = 1, 2$, and $\cup_{\mathcal{Q}_{B,2}(J)} \subset \cup_{\mathcal{Q}_{B,1}(J)} \subset N_{\Delta_B}(J)$.
(2) As $E_{\mathcal{T}''_{B,k}, Y''_{B,k}} \subset E_{L_3}$, we have $|E_{L_3} \cap Q| \gtrsim (\delta/r)^{1-s+2\eta} d_{B,1}(J)^{-1} (r/\Delta_B)|Q|$ for each $Q \in \mathcal{Q}_{B,1}(J)$.
(3) For each $Q \in \mathcal{Q}_{B,2}(J)$, Q contains $\gtrsim (\delta/r)^{-s-\eta} d_{B,2}(J)^{-1}$ many δ-balls in $Y'''_{B,k}(J)$.

Let us summarize what we have so far: $(L_6, Y_6)_\delta$ is a refinement of $(L_2, Y_2)_\delta$, and for $(L_6, Y_6)_\delta$, either one of the following must happen:

A. For each $x \in E_{L_6}$,

$$\#L_6(x) \leqslant \sum_{k \in \mathcal{K}_B(\mathbf{X})} \mu_{B,k} \lessapprox \#\mathcal{K}_B(\mathbf{X}) \cdot \mu_{B,k} \lessapprox \mu_3 \cdot (\delta/r)^{(s-t)/2}. \tag{3.76}$$

B. For each $B \in \mathcal{B}_{r,6}$, we have

$$|E_{L_6} \cap B| \geqslant \max_{k \in \mathcal{K}_B(\mathbf{X})} |E_{\mathcal{T}'''_{B,k}, Y'''_{B,k}}| \gtrapprox (\delta/r)^{\varepsilon^2+3\eta}(\delta/r)^{2-(t+3s)/2}|B|. \tag{3.77}$$

C. For all $B \in \mathcal{B}_{r,6}$, there exists a scale $\Delta_B \geqslant r(\delta/r)^{1-\sqrt{\eta}}$ such that for all $\ell \in L_6$, there are two numbers $d_{B,1}(\ell), d_{B,2}(\ell)$ and two families of disjoint Δ_B-balls $\mathcal{Q}_{B,1}(\ell), \mathcal{Q}_{B,2}(\ell)$, and the following is true:
   (a) $\#\mathcal{Q}_{B,j}(\ell) = d_{B,j}(\ell), j = 1, 2$, and $\cup_{\mathcal{Q}_{B,2}(\ell)} \subset \cup_{\mathcal{Q}_{B,1}(\ell)} \subset N_{\Delta_B}(\ell) \cap B$.
   (b) $|E_{L_3} \cap Q| \gtrsim (\delta/r)^{1-s+2\eta} d_{B,1}(\ell)^{-1}(r/\Delta_B)|Q|$ for each $Q \in \mathcal{Q}_{B,1}(\ell)$.
   (c) For each $Q \in \mathcal{Q}_{B,2}(\ell)$, Q contains $\gtrsim (\delta/r)^{-s-\eta} d_{B,2}(\ell)^{-1}$ many δ-balls in $Y_6(\ell) \cap B$.

We proceed to the next step.

Step 5: Estimate $|E_L|$: Three cases.

Case A. Suppose item **A** happens. By (3.53) and (3.48), we get

$$\#L_6(x) \lessapprox c_\varepsilon^{-1} r^{-\varepsilon} (C\tilde{C}_2)^{\eta^{-2}} \tilde{\lambda}^{-1/2} r^{(1-t)/2} \gamma_{Y_2,t*}^{1/2} \cdot (\delta/r)^{(s-t)/2} \tag{3.78}$$

$$\lessapprox c_\varepsilon^{-1} r^{-\varepsilon} (r/\delta)^{\eta_1} (C\tilde{C}_2)^{\eta^{-2}} \lambda^{-1/2} \delta^{(1-t)/2} \gamma_{Y_2,t*}^{1/2}. \tag{3.79}$$

Since $(L_6, Y_6)_\delta$ is a refinement of $(L, Y)_\delta$ and since $\gamma_{Y_2,t*} \leqslant \gamma_{Y,t*}$, we thus have,

$$|E_L| \geqslant |E_{L_6}| \gtrapprox c_\varepsilon r^\varepsilon (r/\delta)^{-\eta_1} (C\tilde{C}_2)^{-\eta^{-2}} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y,t*}^{-1/2} \sum_{\ell \in L_6} |Y_6(\ell)| \tag{3.80}$$

$$\geqslant c_\varepsilon \delta^\varepsilon (C\tilde{C}_2)^{-\eta^{-2}} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y,t*}^{-1/2} \sum_{\ell \in L} |Y(\ell)|. \tag{3.81}$$

The last inequality follows from the fact $\eta_1 \leqslant \varepsilon$. This concludes (3.34).

Case B. Suppose item **B** happens. Then for each $B \in \mathcal{B}_{r,6}$,

$$|E_L \cap B| \geqslant |E_{L_6} \cap B| \gtrapprox (\delta/r)^{\varepsilon^2+3\eta}(\delta/r)^{2-(t+3s)/2}|B|. \tag{3.82}$$

Let L_7 be the set of lines $\ell \in L_6$ such that $|Y_6(\ell)| \gtrapprox |Y_2(\ell)| \approx \lambda\delta$. Since $(L_6, Y_6)_\delta$ is a refinement of $(L_2, Y_2)_\delta$, $\#L_7 \gtrapprox \#L_2$. For each $\ell \in L_7$, let $Y_7(\ell)$ be a uniform refinement of $Y_6(\ell)$, so $|Y_7(\ell)| \approx \lambda\delta$. Note that $(L_7, Y_7)_\delta$ is a refinement of $(L_2, Y_2)_\delta$.

For each $\ell \in L_7$, let $\mathcal{B}_{r,6}(\ell) \subset \mathcal{B}_{r,6}$ be so that $Y_7(\ell) \cap B \neq \varnothing$ for all $B \in \mathcal{B}_{r,6}(\ell)$. Now define a new shading

$$\tilde{Y}_8(\ell) = \cup_{\mathcal{B}_{r,6}(\ell)} \cap N_r(\ell) \tag{3.83}$$

by r-balls. Since $|Y_7(\ell) \cap B| \lesssim (\delta/r)^{1-s-\eta_1}|N_\delta(\ell) \cap B|$ for each $B \in \mathcal{B}_{r,6}(\ell)$ (from the setup before Step 1), we have (recall (3.48))

$$|\tilde{Y}_8(\ell)|/|N_r(\ell)| \gtrsim \delta^{-1}|Y_7(\ell)| \cdot (r/\delta)^{1-s-\eta_1} \gtrapprox \tilde{\lambda}. \tag{3.84}$$

Moreover, since $\gamma_{\tilde{Y}_2,t*} \lesssim \gamma_{Y_2,t*}$, we have $\gamma_{\tilde{Y}_8,t*} \lesssim \gamma_{Y_2,t*}$

By Lemma 2.7, there exists $\tilde{L}_8 \subset L_7$ such that $(r^t \# \tilde{L}_8) \gtrapprox (\delta^t \# L_7)$, and $\tilde{L}_8$ is a Katz–Tao (r, t)-set. For all $\ell \in L_7$, since $Y_7(\ell)$ is a refinement of $Y_2(\ell)$ and since $Y_2(\ell)$ is a $(\delta, \varepsilon^2, CC_2; \rho)$-set for some $C_2 \lessapprox 1$, $Y_7(\ell)$ is a $(\delta, \varepsilon^2, CC_7; \rho)$-set for some $C_7 \lessapprox 1$. Since $Y_7(\ell)$ is uniform, $\tilde{Y}_8(\ell)$ is a $(r, \varepsilon^2, CC_8; \max\{\rho, r\})$-set for some

$C_8 \lessapprox 1$. Clearly, $\max\{\rho, r\} \in [r, r^\eta]$, since $\rho \in [\delta, \delta^\eta]$. Apply Proposition 3.12 at scale r to $(\tilde{L}_8, \tilde{Y}_8)_r$ so that

$$|E_{\tilde{L}_8}| \geqslant c_\varepsilon r^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} \tilde{\lambda}^{1/2} r^{(t-1)/2} \gamma_{\tilde{Y}_8,t*}^{-1/2} \sum_{\ell \in \tilde{L}_8} |\tilde{Y}_8(\ell)| \tag{3.85}$$

$$\gtrsim c_\varepsilon r^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} \tilde{\lambda}^{3/2} r^{(t-1)/2} \gamma_{\tilde{Y}_8,t*}^{-1/2} (r\#\tilde{L}_8). \tag{3.86}$$

Since $(r^t\#\tilde{L}_8) \lessapprox (\delta^t\#L_7)$, since $\gamma_{\tilde{Y}_8,t*} \lesssim \gamma_{Y_2,t*}$, and since $|Y_7(\ell)| \approx \lambda\delta$ for all $\ell \in L_7$, the above gives

$$\begin{aligned} |E_{\tilde{L}_8}| &\gtrsim c_\varepsilon r^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} \tilde{\lambda}^{3/2} r^{(t-1)/2} \gamma_{Y_2,t*}^{-1/2} \cdot \lambda^{-1} (r/\delta)^{1-t} \sum_{\ell \in L_7} |Y_7(\ell)| \\ &\gtrsim c_\varepsilon r^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} \lambda^{1/2} r^{(t-1)/2} \gamma_{Y_2,t*}^{-1/2} (r/\delta)^{3(1-s-\eta_1)/2} (r/\delta)^{1-t} \sum_{\ell \in L_7} |Y_7(\ell)|. \end{aligned} \tag{3.87}$$

Therefore, by (3.82), (3.87) implies

$$|E_L| \gtrapprox \min_{B \in \mathcal{B}_{r,6}} \frac{|E_L \cap B|}{|B|} \cdot |E_{\tilde{L}_8}| \gtrapprox (\delta/r)^{\varepsilon^2+3\eta} (\delta/r)^{2-(t+3s)/2} |E_{\tilde{L}_8}| \tag{3.88}$$

$$\gtrapprox (\delta/r)^{\varepsilon^2+3\eta+3\eta_1/2} \cdot c_\varepsilon r^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y_2,t*}^{-1/2} \sum_{\ell \in L_7} |Y_7(\ell)| \tag{3.89}$$

$$\gtrapprox (\delta/r)^{\varepsilon^2+3\eta+2\eta_1-\varepsilon} \cdot c_\varepsilon \delta^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y,t*}^{-1/2} \sum_{\ell \in L} |Y(\ell)|. \tag{3.90}$$

Since $\eta_1 \leqslant \eta \leqslant \varepsilon^2$, we have $(\delta/r)^{\varepsilon^2+3\eta+2\eta_1-\varepsilon} \geqslant 1$. This proves (3.34).

Case C. Suppose item **C** happens. Then for each $B \in \mathcal{B}_{r,6}$, there exists a scale $\Delta_B \geqslant r(\delta/r)^{1-\sqrt{\eta}}$. Note that $Y_6(\ell) \subset \cup_{\mathcal{B}_{r,6}}$for any $\ell \in L_6$. Since $(L_6, Y_6)_\delta$ is a refinement of $(L_2, Y_2)_\delta$, we have

$$\int_{E_{L_2}} \#L_2(x) \lessapprox \int_{\cup_{\mathcal{B}_{r,6}}} \#L_6(x). \tag{3.91}$$

By dyadic pigeonholing, there exist $\mathcal{B}'_{r,6} \subset \mathcal{B}_{r,6}$ and $\Delta \geqslant r(\delta/r)^{1-\sqrt{\eta}}$ so that

(1) $\int_{E_{L_2}} \#L_2(x) \lessapprox \int_{\cup_{\mathcal{B}'_{r,6}}} \#L_6(x)$.
(2) For all $B \in \mathcal{B}'_{r,6}$ and all $\ell \in L_6$, there are two numbers $d_{B,1}(\ell), d_{B,2}(\ell)$ and two families of disjoint Δ-balls $\mathcal{Q}_{B,1}(\ell), \mathcal{Q}_{B,2}(\ell)$ such that:
   (a) $\#\mathcal{Q}_{B,j}(\ell) = d_{B,j}(\ell), j = 1, 2$, and $\cup_{\mathcal{Q}_{B,2}(\ell)} \subset \cup_{\mathcal{Q}_{B,1}(\ell)} \subset N_\Delta(\ell) \cap B$.
   (b) $|E_{L_3} \cap Q| \gtrsim (\delta/r)^{1-s+2\eta} d_{B,1}(\ell)^{-1} (r/\Delta)|Q|$ for each $Q \in \mathcal{Q}_{B,1}(\ell)$.
   (c) For each $Q \in \mathcal{Q}_{B,2}(\ell)$, Q contains $\gtrsim (\delta/r)^{-s-\eta} d_{B,2}(\ell)^{-1}$ many δ-balls in $Y_6(\ell) \cap B$.

Thus, if denoting $Y'_6(\ell) = Y_6(\ell) \cap \cup_{\mathcal{B}'_{r,6}}$ for all $\ell \in L_6$, then $(L_6, Y'_6)_\delta$ is a refinement of $(L_2, Y_2)_\delta$. Let $(L_7, Y_7)_\delta$ be a refinement of $(L_6, Y'_6)_\delta$ such that $|Y_7(\ell)| \lessapprox |Y_2(\ell)|$ and $Y_7(\ell)$ is uniform for all $\ell \in L_7$. Hence $|Y_7(\ell)| \approx \lambda\delta$ for all $\ell \in L_7$.

For each $\ell \in L_7$, let $\mathcal{B}'_{r,6}(\ell) \subset \mathcal{B}'_{r,6}$ be so that $Y_7(\ell) \cap B \neq \varnothing$ for any $B \in \mathcal{B}'_{r,6}(\ell)$. By dyadic pigeonholing on $\{d_{B,1}(\ell), d_{B,2}(\ell) : B \in \mathcal{B}'_{r,6}(\ell)\}$, there is a set $\mathcal{B}''_{r,6}(\ell) \subset \mathcal{B}'_{r,6}(\ell)$ and two numbers $d_1(\ell), d_2(\ell)$ such that

(1) For all $B \in \mathcal{B}''_{6,r}(\ell)$, $d_{B,j}(\ell) \sim d_j(\ell), j = 1, 2$.
(2) $|Y'_7(\ell)| \lessapprox |Y_7(\ell)| \approx \lambda\delta$, where $Y'_7(\ell) := Y_7(\ell) \cap \cup_{\mathcal{B}''_{6,r}(\ell)}$.

Thus, $(L_7, Y_7')_\delta$ is a refinement of $(L_6, Y_6)_\delta$. By dyadic pigeonholing again, there is a refinement L_8 of L_7 and two uniform numbers d_1, d_2 such that for all $\ell \in L_8$, $Y_7'(\ell)$ is $\gtrapprox$ 1-refinement of $Y_7(\ell)$, and $d_j(\ell) \sim d_j, j = 1, 2$. The configuration (L_8, Y_7', d_1, d_2) satisfies that for all $\ell \in L_8$, all $B \in \mathcal{B}''_{6,r}(\ell)$, we have the following:

(1) $\cup_{\mathcal{Q}_{B,2}(\ell)} \subset \cup_{\mathcal{Q}_{B,1}(\ell)} \subset N_\Delta(\ell) \cap B$.
(2) $|E_{L_3} \cap Q| \gtrsim (\delta/r)^{1-s+2\eta} d_1^{-1} (r/\Delta)|Q|$ for each $Q \in \mathcal{Q}_{B,1}(\ell)$.
(3) For each $Q \in \mathcal{Q}_{B,2}(\ell)$, Q contains $\gtrsim (\delta/r)^{-s-\eta} d_2^{-1}$ many δ-balls in $Y_7(\ell) \cap B$.

Now for each $\ell \in L_8$, we define three new shadings

(3.92)
$$Y_8(\ell) = Y_7'(\ell) \bigcap \cup_{\mathcal{B}'''_{r,6}(\ell)},\ \tilde{Y}_8(\ell) = \bigcup_{B \in \mathcal{B}'''_{r,6}(\ell)} \cup_{\mathcal{Q}_{B,1}(\ell)},\ \tilde{Y}_8'(\ell) = \bigcup_{B \in \mathcal{B}''_{r,6}(\ell)} \cup_{\mathcal{Q}_{B,2}(\ell)}.$$

Since $d_1|\tilde{Y}_8'(\ell)| \sim d_2|\tilde{Y}_8(\ell)|$, we have $\gamma_{\tilde{Y}_8, t^*} \lesssim (d_1/d_2)\gamma_{\tilde{Y}_8', t^*}$. Also, by Lemma 2.22, we know $\gamma_{\tilde{Y}_8', t^*} \lesssim (\delta/\Delta)^{-t^*}(\delta/r)^{s+\eta} d_2 \gamma_{Y_7, t^*}$. Thus, we have

$$\gamma_{\tilde{Y}_8, t^*} \lesssim (\delta/\Delta)^{-t^*}(\delta/r)^{s+\eta} d_1 \gamma_{Y_7, t^*}. \tag{3.93}$$

Define $d := (\delta/r)^{-s-\eta}(\delta/\Delta)d_1^{-1}$ to simplify notation. Then

$$\gamma_{\tilde{Y}_8, t^*} \lesssim (\delta/\Delta)^{1-t^*} d^{-1} \gamma_{Y_7, t^*}. \tag{3.94}$$

Moreover, for all $\ell \in L_8$, all $B \in \mathcal{B}''_{6,r}(\ell)$, and all $Q \in \mathcal{Q}_{B,1}(\ell)$, we have

$$|E_{L_3} \cap Q| \gtrsim (\delta/r)^{3\eta} d|Q|. \tag{3.95}$$

Since for each r-ball $B \in \mathcal{B}'_{r,6}$, $|Y_7(\ell) \cap B| \lesssim |Y_2(\ell) \cap B| \lesssim (\delta/r)^{1-s-\eta_1}|N_\delta(\ell) \cap B|$, and since $|Y_7(\ell)| \approx |Y_7'(\ell)| \approx \lambda\delta$, we know that $\#\mathcal{B}''_{r,6}(\ell) \gtrapprox (\delta r)^{-1}|Y_7(\ell)|(r/\delta)^{1-s-\eta_1}$. Therefore, $\tilde{Y}_8(\ell)$ contains $\gtrapprox d_1 \cdot (\delta r)^{-1}|Y_7(\ell)|(r/\delta)^{1-s-\eta_1} = (\delta/r)^{-s-\eta}(\delta/\Delta)d^{-1} \cdot (\delta r)^{-1}\lambda\delta(r/\delta)^{1-s-\eta_1}$ many Δ-balls. Simplify the equation to get that $\tilde{Y}_8(\ell)$ contains $\gtrapprox (\delta/r)^{\eta_1-\eta} d^{-1}\Delta^{-1}\lambda$ many Δ-balls. This gives

$$|\tilde{Y}_8(\ell)| \gtrapprox (\delta/r)^{\eta_1-\eta} d^{-1}\Delta^{-1}\lambda \cdot \Delta^2 \sim (\delta/r)^{\eta_1-\eta} d^{-1}\lambda\Delta. \tag{3.96}$$

In particular, $\tilde{Y}_8$ is $\gtrapprox (\delta/r)^{\eta_1-\eta} d^{-1}\lambda$-dense.

By Lemma 2.7, there exists $\tilde{L}_8 \subset L_8$ such that $(\Delta^t \# \tilde{L}_8) \gtrapprox (\delta^t \# L_8)$, and $\tilde{L}_8$ is a Katz–Tao (Δ, t)-set. Since L_8 is a refinement of L_7, $(\Delta^t \# \tilde{L}_8) \gtrapprox (\delta^t \# L_7)$. Now we are going to use the assumption $r \leqslant \delta^\eta$. For all $\ell \in L_8$, since $Y_8(\ell)$ is a refinement of $Y_2(\ell)$ and since $Y_2(\ell)$ is a $(\delta, \varepsilon^2, CC_2; \rho)$-set for some $C_2 \lessapprox 1$, $Y_8(\ell)$ is a $(\delta, \varepsilon^2, CC_8; \rho)$-set for some $C_8 \lessapprox 1$. Since $Y_7(\ell)$ is uniform, by the definition of Y_8 in (3.92), $(Y_8(\ell))_r$ is a $(r, \varepsilon^2, C\tilde{C}_8; \max\{\rho, r\})$-set for some $\tilde{C}_8 \lessapprox 1$. Finally, since $|\tilde{Y}_8(\ell) \cap J|_\Delta \sim d$ for all $J \subset (\tilde{Y}_8(\ell))_r$ and by (3.92), $\tilde{Y}_8(\ell)$ is a $(\Delta, \varepsilon^2, C\tilde{C}_8; \max\{\rho, r\})$-set. Note that $\max\{\rho, r\} \in [\Delta, \Delta^\eta]$, since $\rho \in [\delta, \delta^\eta]$ and since $r \leqslant \delta^\eta$. Apply induction at scale Δ to $(\tilde{L}_8, \tilde{Y}_8)_\Delta$ so that

$$\begin{aligned}
|E_{\tilde{L}_8}| &\gtrapprox c_\varepsilon \Delta^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} (\delta/r)^{(\eta_1-\eta)/2} d^{-1/2} \lambda^{1/2} \Delta^{(t-1)/2} \gamma_{\tilde{Y}_8, t^*}^{-1/2} \sum_{\ell \in \tilde{L}_8} |\tilde{Y}_8(\ell)| \\
&\gtrapprox c_\varepsilon \Delta^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} (\delta/r)^{3(\eta_1-\eta)/2} d^{-1/2} \lambda^{1/2} \Delta^{(t-1)/2} \gamma_{\tilde{Y}_8, t^*}^{-1/2} d^{-1} (\delta/\Delta)^{t-1} \sum_{\ell \in L_7} |Y_7(\ell)| \\
&\gtrapprox c_\varepsilon \Delta^\varepsilon (C\tilde{C}_8)^{-\eta^{-2}} (\delta/r)^{3(\eta_1-\eta)/2} \lambda^{1/2} \delta^{(t-1)/2} \gamma_{Y_7, t^*}^{-1/2} d^{-1} \sum_{\ell \in L_7} |Y_7(\ell)|.
\end{aligned}$$

In the last inequality, we use (3.94) (note that when $t<1$, $(\delta/\Delta)^{1-t^*}=(\delta/\Delta)^{1-t}$).

Recall (3.95) that for all Δ-ball $Q\subset E_{\tilde{L}_8}$, $|E_{L_3}\cap Q|\gtrsim d(\delta/r)^{3\eta}|Q|$. Thus, since $(L_7,Y_7)_\delta$ is a refinement of $(L,Y)_\delta$, since $\gamma_{Y_7,t^*}\leqslant\gamma_{Y,t^*}$, and since $\eta_1\leqslant\eta$, we get

$$|E_L|\geqslant d(\delta/r)^{3\eta}|E_{\tilde{L}_8}|\gtrapprox c_\varepsilon\Delta^\varepsilon(C\tilde{C}_8)^{-\eta^{-2}}(\delta/r)^{3\eta}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y_7,t^*}^{-1/2}\sum_{\ell\in L}|Y(\ell)| \tag{3.97}$$

$$\gtrapprox(\delta/r)^{3\eta}(\Delta/\delta)^\varepsilon\cdot c_\varepsilon\delta^\varepsilon(C\tilde{C}_8)^{-\eta^{-2}}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t^*}^{-1/2}\sum_{\ell\in L}|Y(\ell)|. \tag{3.98}$$

Since $\Delta\geqslant r(\delta/r)^{1-\sqrt{\eta}}$, we have $\Delta/\delta\geqslant(r/\delta)^{\sqrt{\eta}}$. Also, since $\eta_1\leqslant\eta\leqslant\varepsilon^{10}$, we get $(\delta/r)^{3\eta}(\Delta/\delta)^{-\varepsilon}\geqslant(r/\delta)^{\varepsilon\sqrt{\eta}-3\eta}\geqslant(r/\delta)^\eta\geqslant1$. Consequently, we have $|E_L|\geqslant c_\varepsilon\delta^\varepsilon(C\tilde{C}_8)^{-\eta^{-2}}\lambda^{1/2}\delta^{(t-1)/2}\gamma_{Y,t^*}^{-1/2}\sum_{\ell\in L}|Y(\ell)|$. This concludes Proposition 3.12. □

References

[DW24] Ciprian Demeter and Hong Wang. Szemerédi-Trotter bounds for tubes and applications. *arXiv:2406.06884*, 2024.

[OS23] Tuomas Orponen and Pablo Shmerkin. Projections, furstenberg sets, and the abc sum-product problem. *arXiv:2301.10199*, 2023.

[RW23] Kevin Ren and Hong Wang. Furstenberg sets estimate in the plane. *arXiv preprint arXiv:2308.08819*, 2023.

[Shm23] Pablo Shmerkin. A non-linear version of Bourgain's projection theorem: Dedicated to the memory of Jean Bourgain. *Journal of the European Mathematical Society (EMS Publishing)*, 25(10), 2023.

[Wol99] Thomas Wolff. Recent work connected with the Kakeya problem. In *Prospects in mathematics (Princeton, NJ, 1996)*, pages 129–162. Amer. Math. Soc., Providence, RI, 1999.

[WW24] Hong Wang and Shukun Wu. Restriction estimates using decoupling theorems and two-ends Furstenberg inequalities. *arXiv:2411.08871*, 2024.

Courant institute of mathematical sciences, New York University
Email address: hw3639@nyu.edu

Department of Mathematics, Indiana University Bloomington
Email address: shukwu@iu.edu